

Learning Metamaterial Eigenmodes with Wavelet-Encoded Fourier Neural Operators

Han Zhang¹, Alexander Ogren², Cynthia Rudin³,
Johann Guilleminot¹, L. Catherine Brinson¹

¹*Department of Mechanical Engineering and Materials Science, Duke University, Durham, NC, USA*

²*Department of Mechanical Engineering, California Institute of Technology, Pasadena, CA, USA*

³*Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA*

Abstract

Machine learning surrogates based on neural operators have shown broad applicability in solving forward PDE problems. However, eigenvalue problems, in which an eigenparameter and one of several valid eigenmodes must be simultaneously solved, remain difficult because standard operator learning formulations assume a unique input–output map. This work demonstrates that Fourier Neural Operators (FNOs), combined with wavelet-based encodings of PDE inputs, can learn and inexpensively predict multiple eigenmodes of the elastic wave equation, corresponding to deformation modes of acoustic waves propagating through arbitrary metamaterial geometries. We provide theoretical motivation and experimental evidence for why wavelet encodings are well matched to the dual spatial–spectral structure of the FNO, enabling deterministic mode selection on both continuous and discretely valued geometries within a single model, and for why prediction accuracy varies with geometric discontinuities. For metamaterial design, the resulting surrogate accelerates the simulation stage of the design cycle by three orders of magnitude relative to finite element analysis on a consumer-grade CPU, while preserving high fidelity. These results also carry broader implications for designing input encodings in other multi-mode PDE solvers based on spectral neural operators.

Keywords: Neural Operator, Metamaterials, Wavelets, Eigenmodes, Machine Learning

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1. Introduction

Acoustic metamaterials are architected materials that enable compact control of sound and elastic waves, including sound attenuation, vibration isolation, and wave focusing, capabilities that are difficult to achieve with homogeneous materials alone [1, 2]. These properties arise from geometric structure in addition to material composition, so the dispersive response of a periodic unit cell is governed by the elastic wave equation and is compactly summarized by phononic band structures obtained from Bloch eigenvalue analysis [3, 4].

Understanding and designing such systems requires computing multiple deformation modes associated with different resonant frequencies and wave behaviors. Traditionally, these modes are obtained through computationally expensive eigenvalue analysis using finite element methods [5]. While FEA is highly reliable, this repeated evaluation can make the simulation stage of design cycles prohibitive for optimization, uncertainty quantification, inverse design, and real-time control [5, 4].

Recent advances in machine learning have introduced neural operators as a promising alternative to traditional numerical solvers. Unlike conventional neural networks that learn mappings between finite-dimensional vectors, neural operators learn mappings between function spaces, allowing them to approximate entire solution operators for families of PDEs [6, 7]. Models such as the Fourier Neural Operator (FNO) have demonstrated remarkable accuracy and computational efficiency across a variety of linear and nonlinear PDE problems while maintaining strong generalization capabilities across discretizations and geometries [8]. Related physics-informed operator architectures likewise seek rapid emulation of parametric PDE solution maps [9]. These developments have generated significant interest in using neural operators as surrogate models for computational physics applications.

Despite this progress, existing neural operator architectures have primarily been developed for PDEs with known governing parameters and unique input-output maps. Eigenvalue PDEs instead require simultaneous recovery of an unknown eigenparameter and one of several valid eigenmodes [10, 4]. In phononic and acoustic metamaterials, prior surrogates have predicted dispersion from material parameters [11], addressed

related eigenvalue problems with convolutional networks [10], accelerated dispersion evaluation with Gaussian processes [12], and used neural operators for transmission-loss spectra [13] or wavevector-conditioned band structures [14]. Deep learning has also been applied to phononic-crystal and metamaterial design [15, 16, 17], including interpretable methods that extract unit-cell design rules for targeted bandgaps [18, 19]. Predicting full Bloch displacement eigenfields while allowing for deterministic multi-mode selection with an FNO remains an unsolved challenge with immense practical benefit.

In this work, we introduce a framework that enables Fourier Neural Operators to solve eigenvalue PDEs associated with elastic wave propagation in acoustic metamaterials. Model inputs are augmented with wavelet-based encodings of PDE parameters that uniquely identify individual eigenmodes while remaining compatible with the FNO architecture. We demonstrate that these encodings allow a single neural operator to learn multiple valid solutions corresponding to distinct deformation modes and to reproduce those modes deterministically on demand. We further show that conventional spatial encodings and purely spectral encodings do not provide the same capability, highlighting the importance of the proposed wavelet representation. We also evaluate the same model on continuous- and discrete-valued geometries, recovering both displacement fields and eigenfrequencies, and examine how geometric discontinuities interact with the truncated spectral structure of the FNO.

The contributions of this paper are threefold:

1. We demonstrate that a single Fourier Neural Operator can learn multiple eigenmodes across continuous and discrete metamaterial geometries, recovering both eigenvectors (displacement fields) and eigenvalues (eigenfrequencies).
2. We introduce wavelet encodings of modal parameters and explain theoretically why their spatial-spectral structure enables deterministic mode selection in FNOs.
3. We show that prediction accuracy degrades with geometric discontinuities and interpret this through spectral truncation, with implications for applying FNOs to problems with sharp interfaces.

Together, these results extend neural operators beyond conventional forward PDE problems and provide a pathway to greatly accelerate acoustic metamaterial simulation and design relative to repeated finite element eigenvalue analysis.

2. Methodology

This section describes the methods used to generate data and train Fourier Neural Operators for acoustic metamaterial eigenmode prediction. We first define the metamaterial design space and the Bloch FEA simulations used for supervised labels. We then summarize the FNO architecture and the wavelet encodings of modal parameters. Finally, we describe dataset construction and the training procedure.

2.1. Metamaterial Design Space

To train a surrogate model capable of predicting acoustic eigenmodes, a representative design space of acoustic metamaterials must first be defined. The design space was selected to provide sufficient geometric and material complexity to demonstrate operator learning for eigenvalue PDEs while maintaining computationally tractable finite element simulations for dataset generation.

The dataset is restricted to two-dimensional acoustic metamaterials composed of infinitely repeating square unit cells exhibiting eight-fold symmetry. Each unit cell is discretized onto a 32×32 spatial grid and represented by three material-property channels corresponding to the normalized elastic modulus, density, and Poisson’s ratio. Material properties are normalized to the interval $[0, 1]$ to improve numerical conditioning during training, while the corresponding physical values are recovered using the fixed material constants listed in Table 1. The two constituent materials were chosen to represent a structural steel and a stiff elastomer.

Parameter	$E_{\text{elastomer}}$	E_{steel}	$\rho_{\text{elastomer}}$	ρ_{steel}	$\nu_{\text{elastomer}}$	ν_{steel}
Units	Pa	Pa	kg/m^3	kg/m^3	—	—
Value	1×10^8	2×10^{11}	1200	8000	0.45	0.3

Table 1: Material parameters. The two constituents are a structural steel and a stiff elastomer.

Unit-cell geometries are synthesized by sampling a Gaussian process with a periodic covariance kernel and subsequently enforcing eight-fold symmetry. This procedure produces smoothly varying spatial material distributions while ensuring compatibility with periodically tiled square lattices. The sampled field assumes values between 0 and 1, where 0 represents pure elastomer, 1 represents pure steel, and intermediate values denote an effective mixture of the two materials below the spatial resolution of the discretization.

To investigate the influence of geometry representation on model performance, two classes of unit cells were generated. The first consists of the continuous-valued Gaussian process realizations described above and is referred to as the *continuous geometry* dataset. The second is obtained by thresholding the Gaussian process samples to binary values, producing geometries composed solely of pure elastomer and pure steel. This second collection is referred to as the *discrete geometry* dataset. Each representation comprises approximately one-half of the total dataset.

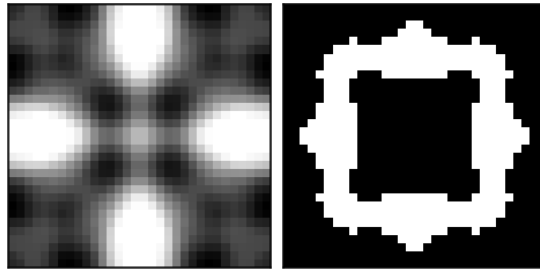


Figure 1: Example of a continuous geometry (left) and a discrete geometry (right) generated using the above process and parameters. White pixels represent the stiffer material, black pixels represent the softer material, and grayscale pixels indicate a homogeneous mix of the two.

2.2. Wave Propagation Simulations

For each metamaterial geometry, wave propagation is simulated by solving the frequency-domain linear elastodynamic equation with finite element analysis (FEA), following the standard Bloch–Floquet treatment of periodic phononic media [4, 3]. After spatial discretization and enforcement of Bloch periodicity, the governing

151 problem takes the generalized eigenvalue form

$$(\mathbf{K}_r(\mathbf{k}) - \omega^2 \mathbf{M}_r(\mathbf{k})) \mathbf{u} = \mathbf{0}, \quad (1)$$

152 where ω is the angular eigenfrequency, $\mathbf{u}$ is the corresponding complex Bloch displace-
 153 ment eigenvector, and $\mathbf{k}$ is the wavevector in the irreducible Brillouin zone (IBZ). The
 154 reduced stiffness and mass matrices $\mathbf{K}_r(\mathbf{k})$ and $\mathbf{M}_r(\mathbf{k})$ are obtained from geometry-
 155 dependent global finite-element matrices $\mathbf{K}$ and $\mathbf{M}$ via a wavevector-dependent Bloch
 156 transformation matrix $\mathbf{T}(\mathbf{k})$,

$$\mathbf{K}_r(\mathbf{k}) = \mathbf{T}(\mathbf{k})^\dagger \mathbf{K} \mathbf{T}(\mathbf{k}), \quad \mathbf{M}_r(\mathbf{k}) = \mathbf{T}(\mathbf{k})^\dagger \mathbf{M} \mathbf{T}(\mathbf{k}), \quad (2)$$

157 with $(\cdot)^\dagger$ the Hermitian transpose. Here $\mathbf{T}(\mathbf{k})$ maps the full mesh degrees of freedom
 158 to an independent set while applying phase factors $e^{i\mathbf{k}\cdot\mathbf{R}}$ across periodic faces of the
 159 unit cell. The matrices $\mathbf{K}$ and $\mathbf{M}$ themselves depend only on the material field and
 160 are assembled once per geometry. These intermediate matrices are described briefly
 161 so the reader can compare the explicit FEA pipeline (Figure 2) with the Fourier
 162 neural operator architecture in Section 2.3 below (they are not inputs to the neural
 163 operator).

164 The IBZ is discretized into 25 x -direction and 13 y -direction wave numbers,
 165 totaling $25 \times 13 = 325$ unique wavevectors covering the IBZ half-plane. For each
 166 wavevector, the six lowest eigenfrequencies (bands) were computed using a custom
 167 FEA MATLAB script (see Section 4 for implementation details).

168 Each eigenvector $\mathbf{u}$ consists of two complex-valued displacement fields, (u_x, u_y) ,
 169 which are separated into their real and imaginary components to produce four
 170 real-valued output channels, $(u_{x,real}, u_{x,imag}, u_{y,real}, u_{y,imag})$. The corresponding eigen-
 171 frequency ω is encoded as a spatially uniform fifth channel, resulting in a target tensor
 172 of size $5 \times 32 \times 32$. These five-channel targets are associated with the three-channel
 173 material representation from the previous subsection that entered the FEA solve, so
 174 each simulation defines the map

$$\mathbb{R}^{3 \times 32 \times 32} \rightarrow \mathbb{R}^{5 \times 32 \times 32}.$$

175 The three-channel inputs actually ingested by the Fourier neural operator are as-
 176 sembled differently: a geometry channel is combined with wavelet encodings of the
 177 wavevector and band index, as described in Sections 2.4 and 2.5. The overall finite
 178 element data-generation procedure is summarized in Figure 2.

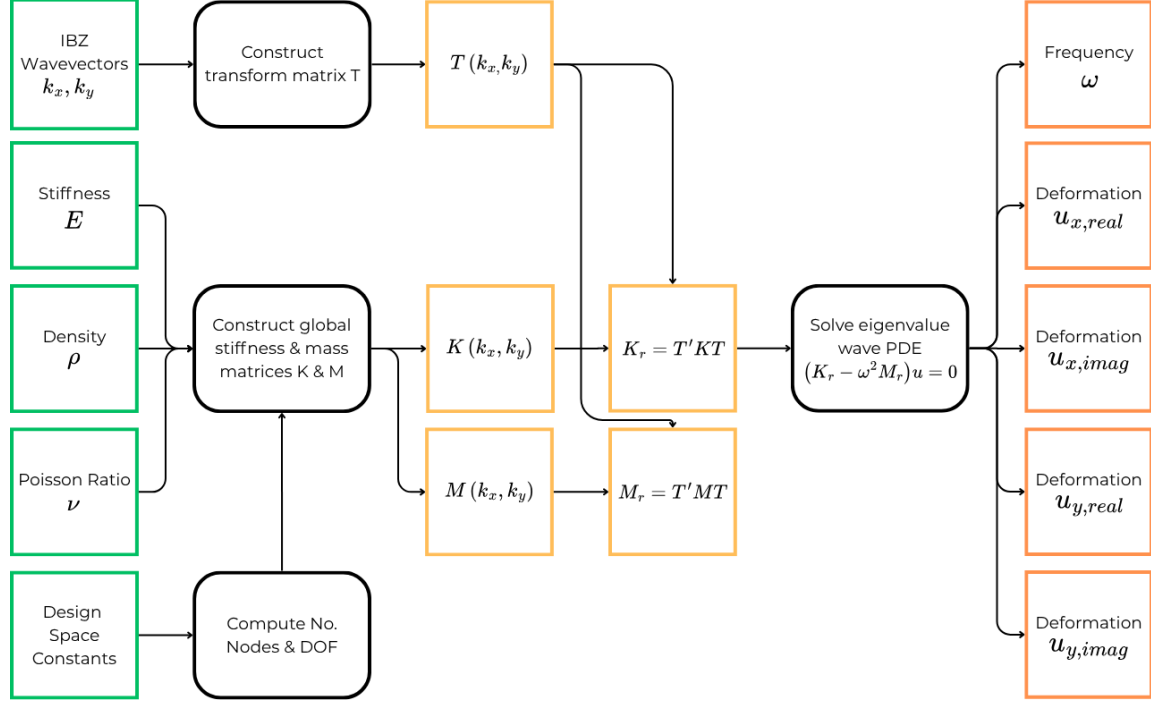


Figure 2: Flowchart of the FEA data generation method. Green squares represent varying inputs to the process, and orange squares represent computed quantities of the FEA process. Black rounded rectangles represent subroutines of the FEA process. The process starts with inputs of defined densities, stiffnesses, and Poisson ratios for each pixel of a metamaterial unit cell geometry (each a 32×32 pixel grid of values ranging from 0 to 1), along with a chosen k_x and k_y wavevector representing the stimulus to the metamaterial. A custom optimized FEA solver takes these inputs and produces intermediate matrices ($\mathbf{K}, \mathbf{M}, \mathbf{T}$), forms the reduced operators ($\mathbf{K}_r, \mathbf{M}_r$), and then solves Eq. (1) for the displacement fields (eigenvectors) and frequencies ω (eigenvalues) for the given inputs.

179 2.3. Fourier Neural Operator

180 The principal theoretical motivation for Fourier Neural Operators is their universal
 181 approximation property over function spaces. Let $\mathcal{A}$ and $\mathcal{U}$ denote Banach spaces of

182 input and output functions, respectively, and let

$$\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U} \quad (3)$$

183 be a continuous nonlinear operator. It has been shown that, given sufficient latent
 184 width and Fourier modes, an FNO can approximate $\mathcal{G}$ arbitrarily well on compact
 185 subsets of $\mathcal{A}$ [6]. Unlike conventional neural networks, which approximate finite-
 186 dimensional functions, FNOs approximate operators between infinite-dimensional
 187 function spaces by learning a sequence of global integral operators parameterized in
 188 the Fourier domain [8].

189 This theoretical framework assumes that both the input and output are continuous
 190 fields of infinite resolution. In practice, however, these fields are sampled on a finite
 191 computational grid. The continuous Fourier transform is therefore replaced by a
 192 discrete Fourier transform (DFT), and only a finite number of Fourier modes are
 193 retained during training [8, 20]. Consequently, the learned model approximates the
 194 continuous solution operator through its discrete representation while preserving the
 195 same spectral operator-learning architecture.

196 For the present work, the operator acts on discretized fields over a 32×32 pixel
 197 spatial grid. The input field consists of three channels: a metamaterial geometry
 198 channel together with wavelet encodings of the stimulatory wavevector and band
 199 index, constructed as detailed in Section 2.4. The output field consists of five
 200 channels corresponding to the predicted eigenvalue, or frequency, and eigenvector, or
 201 displacement field, of the resulting deformation mode. Thus, the discretized learned
 202 operator takes the form

$$\mathcal{G}_\theta : \mathbb{R}^{32 \times 32 \times 3} \rightarrow \mathbb{R}^{32 \times 32 \times 5}. \quad (4)$$

203 The discretized FNO follows the standard lift–transform–project architecture:

- 204 1. **Lifting:** The input field is embedded into a higher-dimensional latent space
 205 through a pointwise lifting operator,

$$v_1(x) = \ell(a(x)), \quad \ell : \mathbb{R}^3 \rightarrow \mathbb{R}^{d_L}. \quad (5)$$

206 **2. Fourier layers:** A sequence of Fourier layers is applied to the latent represen-
 207 tation. Each layer updates the hidden field according to

$$v_{j+1}(x) = \sigma \left(\mathcal{F}^{-1} [R_j(k) \mathcal{F}[v_j](k)] + W_j v_j(x) + b_j \right), \quad (6)$$

208 where $\mathcal{F}$ denotes the discrete Fourier transform, $R_j(k)$ is the learnable spectral
 209 kernel, W_j is a learnable pointwise linear transformation, b_j is a learnable bias
 210 term, and σ is the activation function. GELU was used in this work because it
 211 is smooth and differentiable.

212 **3. Projection:** The final latent representation is projected back to the desired
 213 output dimension through a pointwise projection operator,

$$u(x) = p(v_n(x)), \quad p : \mathbb{R}^{d_L} \rightarrow \mathbb{R}^5. \quad (7)$$

214 FNOs are well suited for acoustic metamaterial simulation because the underlying
 215 physics is governed by spatially distributed wave equations, where the response
 216 at one location depends on the global geometry of the unit cell and the imposed
 217 wavevector. The Fourier layers are designed to use spectral convolutions to efficiently
 218 capture long-range interactions and periodic spatial structure, while the pointwise
 219 transformations preserve local geometric information. Although FNOs are commonly
 220 used as surrogate models for fixed PDE solution operators, the same operator-learning
 221 framework can be extended to eigenvalue problems by conditioning the input on
 222 modal parameters, such as the wavevector and band index, and training the network
 223 to output both the associated eigenvalue and eigenvector field. In this formulation,
 224 the FNO learns a parameterized eigensolution operator mapping geometry and modal
 225 encodings to the frequency and deformation mode of the acoustic unit cell.

226 This process is illustrated graphically in Figure 3. For our experiments, the
 227 NeuralOp Python package was used, which provides FNO model definitions with
 228 customizable numbers of layers, hidden channels, and input and output tensor sizes.
 229 The hyperparameter combinations explored, as well as the optimized final FNO
 230 configuration, are shown in Table S1 in Section S1. These hyperparameters are used

231 alongside the fixed parameters of the problem space: model height 32, model width
 232 32, input channels 3, and output channels 5.

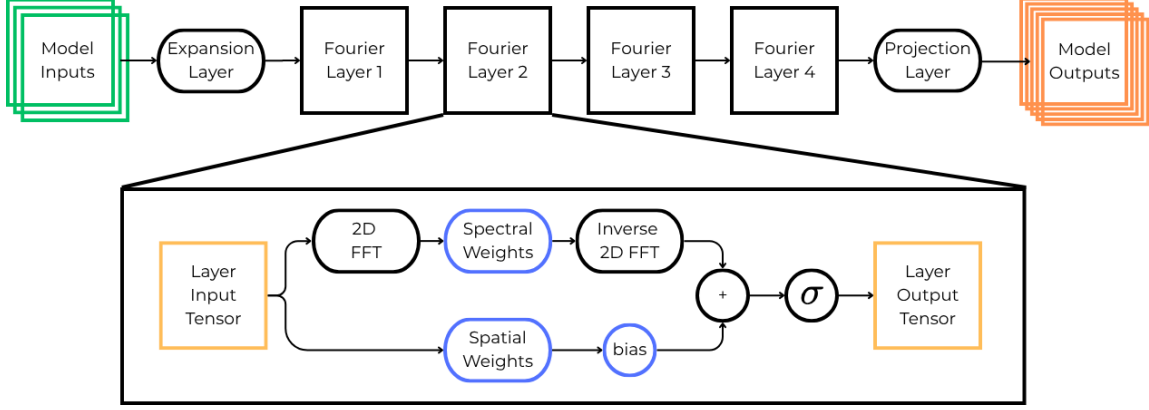


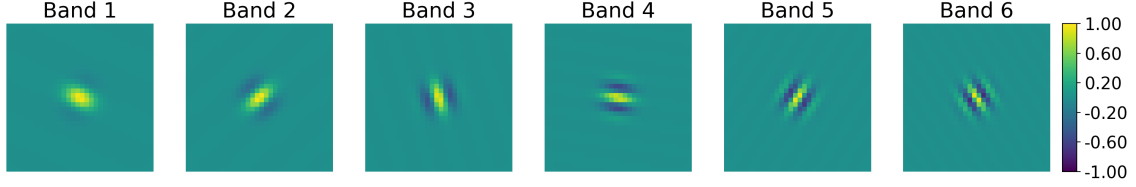
Figure 3: The FNO model architecture used in this project. The input data is first expanded according to the hidden-channels parameter, allowing multiple latent features to be represented between Fourier layers. Each Fourier layer is shown in detail in the expanded view. The data passing through a Fourier layer is processed through two learned branches: a pointwise branch, which applies a learned linear transformation in the spatial representation, and a spectral branch, which applies a fast Fourier transform (FFT), multiplies the retained Fourier modes by learned spectral weights, and then applies an inverse FFT. The outputs of both branches, along with a learnable bias term, are summed before passing through a nonlinear activation function. After passing through four Fourier layers, the latent representation is projected from the hidden-channel dimensionality to the final output dimensions.

233 2.4. Input Wavelet Encoding

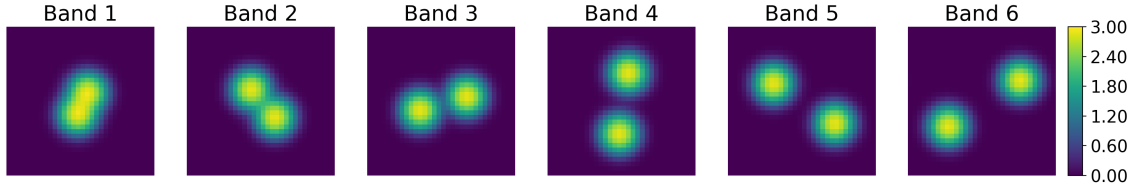
234 To allow the FNO model to toggle between PDE solutions for a given geom-
 235 etry, information must be fed to the model indicating which excitation waveform
 236 (wavevectors) and deformation mode (band) we are looking for a solution to. This
 237 information is given in the form of wavelet embeddings of the wavevector and band
 238 index, which we have found to yield superior results compared with a constant field
 239 embedding (2D matrices of the same value) or a sinusoidal embedding (2D sinusoidal
 240 fields with constants encoded as the sinusoidal frequency). A discussion on why
 241 wavelet encodings are superior is found in Section 3.4.

242 For the wavelet encodings, we use a 1D Gabor wavelet transform to map band
 243 numbers to unique wavelet images (see Algorithm 1), and a 2D Gabor wavelet

244 transform to map x and y wavevectors to another set of unique wavelet images (see
 245 Algorithm 2), following the classical construction of jointly localized space–frequency
 246 atoms [21]. These representations show rich variation both in the spatial domain
 247 and in the Fourier domain, which allows for FNO model weights of both branches to
 248 richly interact with the wavelet representation, leading to better learning outcomes.

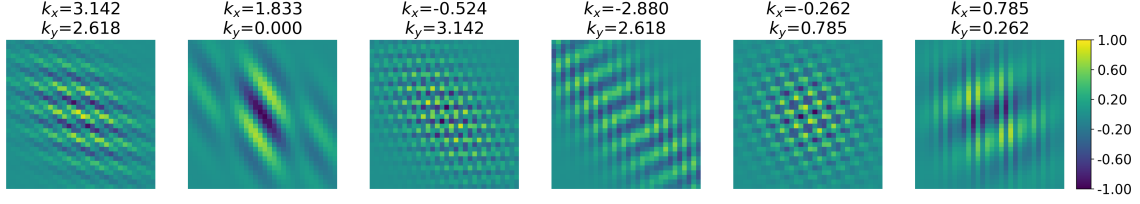


(a) The wavelet spatial encoding for band integer.

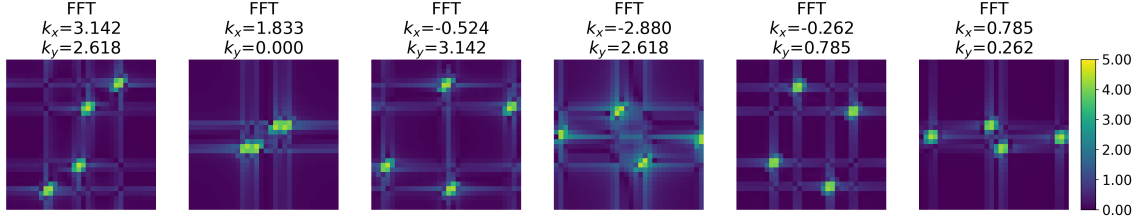


(b) The FFT of the above wavelet spatial encodings.

Figure 4: The wavelet encoding of the band integer (Fig. 4a) and its FFT (Fig. 4b). Each band corresponds to a unique wavelet with representation in both spatial and spectral domains.



(a) The wavelet spatial encoding for x and y wavevectors.



(b) The FFT of the above wavelet spatial encodings.

Figure 5: The wavelet encoding of the x and y wavevectors (Fig. 5a) and its FFT (Fig. 5b). The physical wavevectors are shown in the subplot titles. For each (k_x, k_y) pair there is a unique mapping in both the spatial and spectral domains.

When encoding continuous constants with oscillatory functions on a finite grid, it is important to verify that distinct parameter values do not alias into nearly identical patches [21]. Otherwise the model cannot distinguish neighboring wavevectors, and mode selection collapses. Let $\psi_{\mathbf{k}} \in \mathbb{R}^{S \times S}$ be the 2D wavelet encoding of wavevector $\mathbf{k}$ as described above, and let $\hat{\psi}_{\mathbf{k}} \in \mathbb{C}^{S^2}$ be the flattened discrete Fourier transform of that 2D field (FFT computed on the $S \times S$ array, then reshaped to a vector). Define the mean-centered spectral descriptor by applying the complex modulus and base-10 logarithm entrywise:

$$\mathbf{s}_{\mathbf{k}} = \log_{10}|\hat{\psi}_{\mathbf{k}}| - \text{mean}(\log_{10}|\hat{\psi}_{\mathbf{k}}|)$$

where $|\cdot|$ and $\log_{10}$ act elementwise. Pairwise spectral similarity is then the cosine similarity:

$$S(\mathbf{k}, \mathbf{k}') = \frac{\mathbf{s}_{\mathbf{k}} \cdot \mathbf{s}_{\mathbf{k}'}}{\|\mathbf{s}_{\mathbf{k}}\| \|\mathbf{s}_{\mathbf{k}'}\|}$$

with $\|\cdot\|$ the Euclidean vector norm. Evaluating S over all 325 IBZ wavevectors yields the matrix in Figure 6. The maximum off-diagonal entry is 0.862, well below near-duplicate levels, confirming that the chosen encoding constants introduce no

262 appreciable aliasing across the IBZ grid.

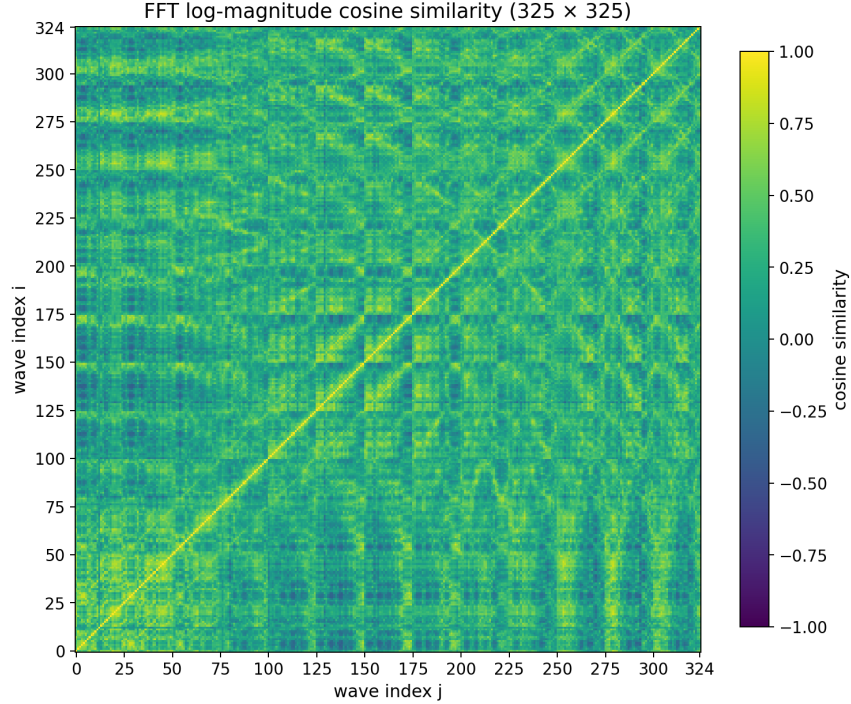


Figure 6: Pairwise cosine similarity of mean-centered $\log_{10} |\text{FFT}|$ spectra for the 325 IBZ wavevector encodings generated by Algorithm 2. The maximum off-diagonal similarity is 0.862, indicating that distinct (k_x, k_y) pairs remain spectrally separable on the 32×32 grid.

263 2.5. Dataset Construction

264 Our full dataset contains 24,000 geometries, randomly generated as described in
 265 prior subsections. Half are discrete-valued, with only 0s or 1s in each pixel location
 266 representing one of the two materials, while the other half are continuous-valued, with
 267 any value in $[0, 1]$ at each pixel representing a blended material. Each geometry has
 268 six bands and 325 wavevectors per band (a mesh grid of the IBZ half-plane), resulting
 269 in $24000 \times 6 \times 325 = 46,800,000$ sample pairs of inputs with shape $(3 \times 32 \times 32)$ and
 270 outputs of shape $(5 \times 32 \times 32)$. The data are stored as float16 PyTorch tensors.

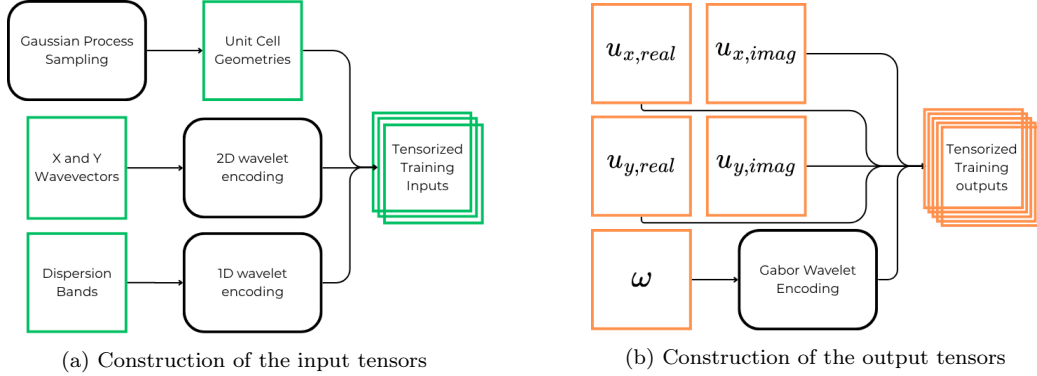


Figure 7: Subfigures a and b show the processes by which input data from Section 2.1 and output data from Section 2.2 are tensorized for ingestion by the FNO model. Note the wavelet embedding steps applied to fundamentally scalar quantities.

2.6. Training Procedure

The Fourier Neural Operator was trained using supervised learning on paired inputs and outputs described in Sections 2.5 and 2.4. Each training sample consists of a $3 \times 32 \times 32$ input tensor (geometry and wavelet embeddings) and a $5 \times 32 \times 32$ output tensor (displacement field components and eigenfrequency).

Five loss functions were evaluated as training objectives: mean absolute error (MAE), mean squared error (MSE), normalized MAE (NMAE), normalized MSE (NMSE), and structural similarity index (SSIM). Losses were aggregated across all five output channels. NMAE yielded the best overall performance and was used for all reported results. Its definition is given below. Details on the other loss functions are provided in the Supplementary Information.

$$\mathcal{L}_{\text{NMAE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 \frac{|\hat{u}_c(i, j) - u_c(i, j)|}{|u_c(i, j)| + \varepsilon}$$

where $H = W = 32$ (pixels), ε is a small stabilizer, and $c \in \{0, 1, 2, 3, 4\}$ corresponds to the 5 output channels: eigenfrequency, followed by real and imaginary components of x and y displacements.

For training the FNO model, combinations of model layers, hidden channels, activation function, loss function, optimizer, and training hyperparameters were

evaluated. The AdamW optimizer [22] achieved the best performance compared with Adam [23], other momentum-based optimizers, and SGD. The candidate settings and selected values are reported in Table S1 in Section S1, with the selected values italicized.

The selected architecture uses four Fourier layers, 128 hidden channels, and GELU activations. Increasing the depth or width beyond these values did not appreciably improve validation performance and increased both training cost and the tendency to overfit. The selected training configuration uses NMAE throughout, AdamW optimization, a learning rate of 2×10^{-3} , a batch size of 520, and 12 epochs. All results reported in this article were generated using this configuration.

Final model performance did not appreciably change with modest variations in scheduler step size or batch size, so these parameters were fixed at 1 and 520, respectively. Learning rate and time to convergence were inversely related: higher learning rates accelerated convergence but produced less stable optimization. In particular, learning rates above approximately 10^{-2} occasionally caused an abrupt, irreversible loss spikes, after which training plateaued. We therefore selected a learning rate of 2×10^{-3} , which provided the best balance between convergence speed and stability.

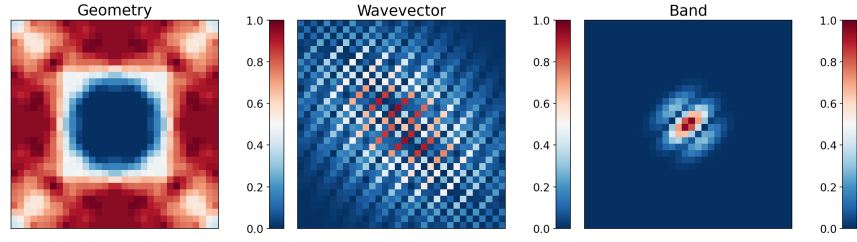
3. Results and Discussion

We organize results as follows. We first assess Bloch displacement predictions on continuous and discrete geometries, then eigenfrequency and dispersion accuracy, then how discrete prediction error tracks material interface length. Finally, we compare wavelet, sinusoidal, and uniform encodings under matched training settings.

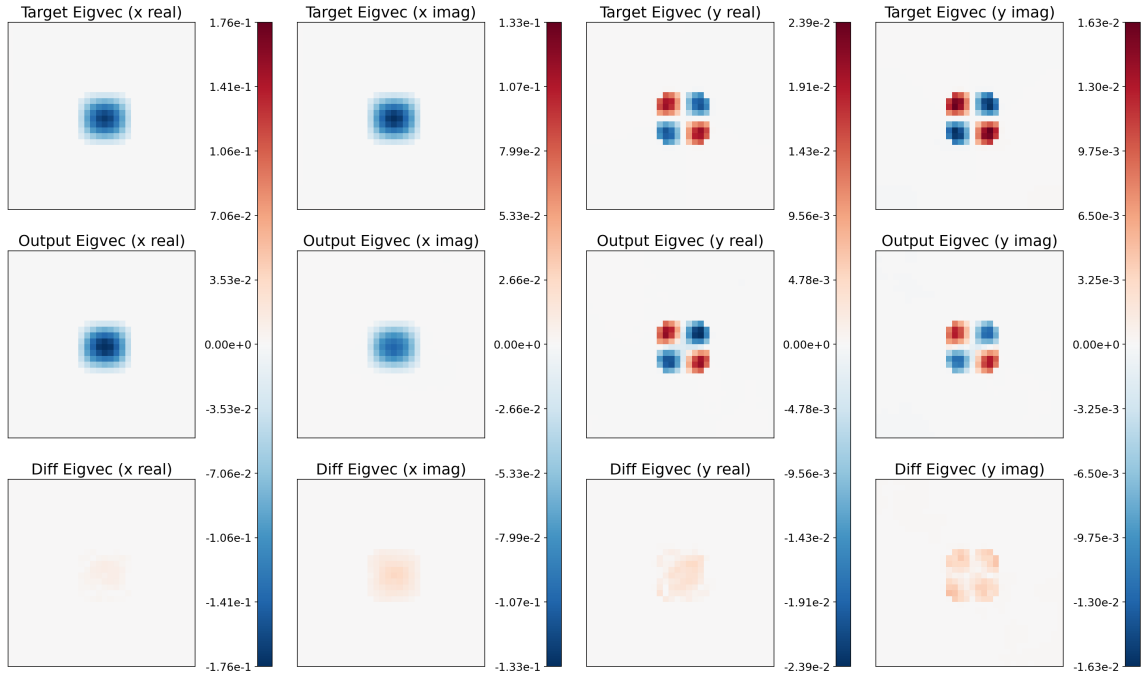
3.1. Displacement Field Prediction

We first evaluate the trained Fourier Neural Operator on its predictions of the Bloch displacement fields (u_x, u_y) . For each unseen test sample, the prediction error is measured by the normalized mean absolute error (NMAE) over the four real-valued displacement channels, $(u_{x,\text{real}}, u_{x,\text{imag}}, u_{y,\text{real}}, u_{y,\text{imag}})$. Samples are ranked by performance, defined inversely with NMAE, so larger losses correspond to lower

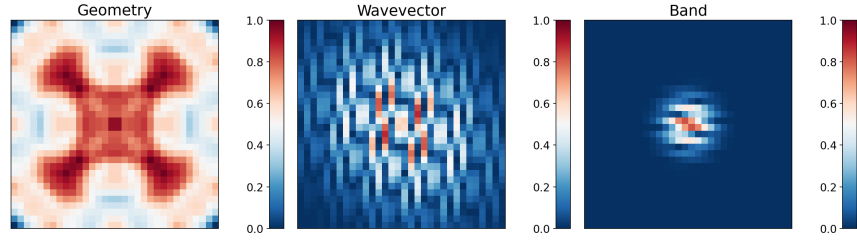
316 performance percentiles. We select representative cases at the 25th, 50th, and 75th
317 performance percentiles, corresponding to relatively weak, median, and relatively
318 strong predictions. The same percentile comparison is shown for both the binarized
319 and continuous material datasets, allowing direct visual assessment of how well the
320 model recovers the spatial structure of the displacement modes across contrasting
321 geometry regimes.



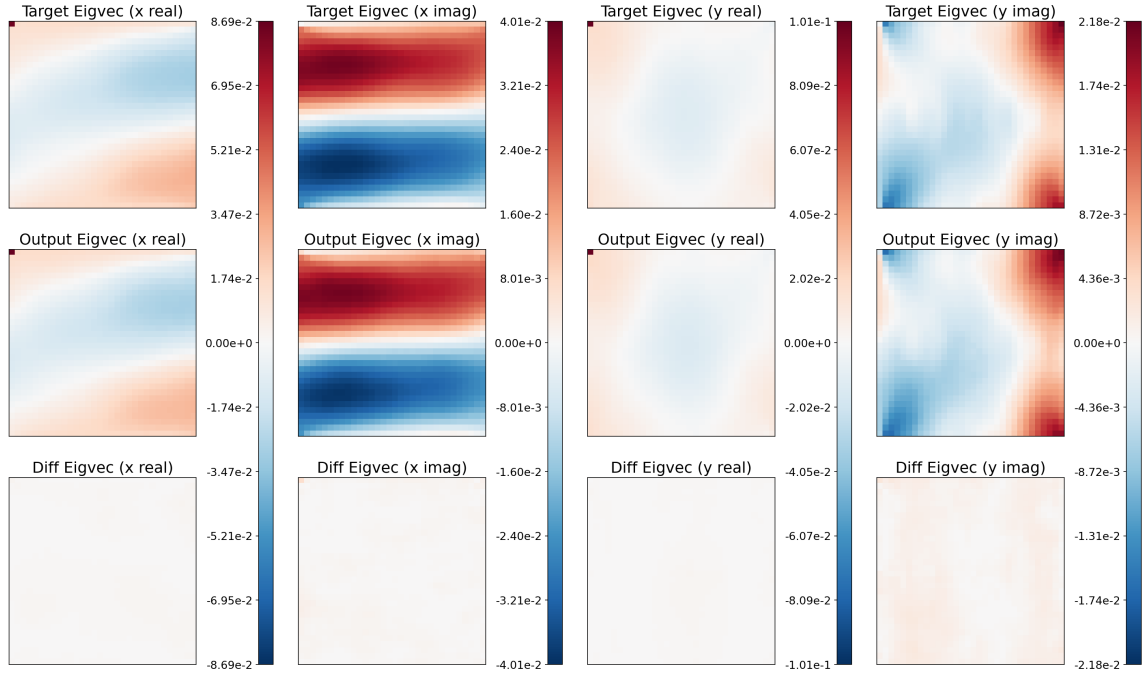
(a) Input sample at the 25th percentile of performance for unseen continuous-valued geometries.



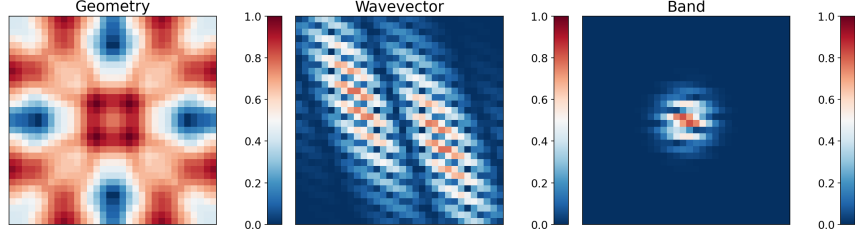
(b) Output predictions at the 25th percentile of performance for unseen continuous-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. At the 25th percentile, prediction outcomes are already fairly good, with accurate representations of scale and structure.



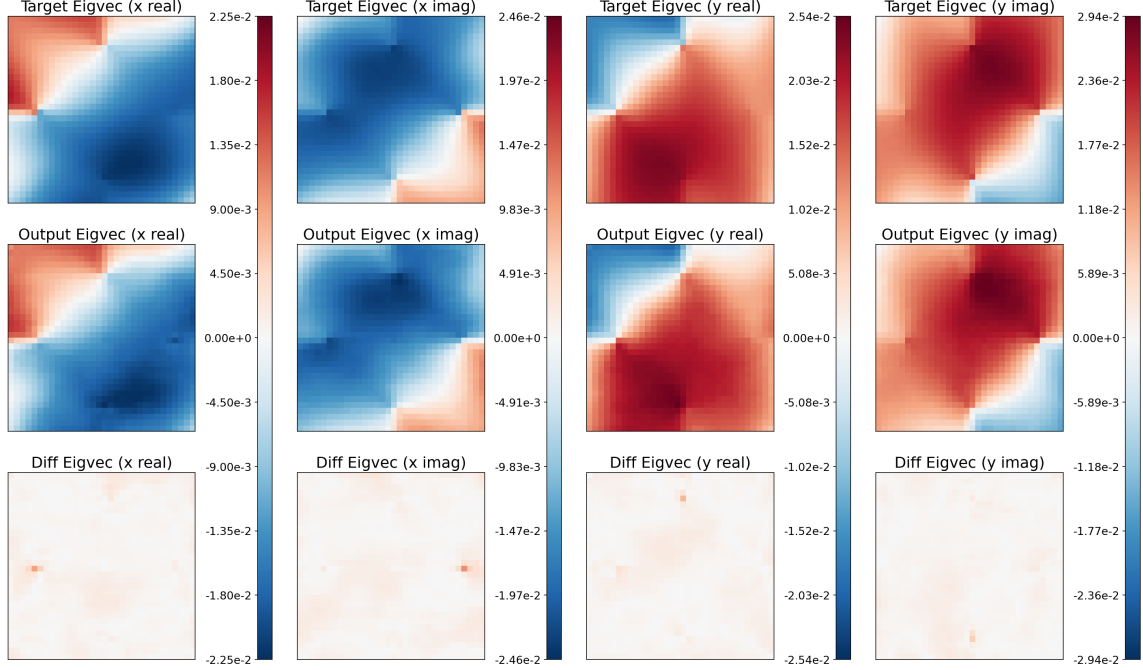
(a) Input sample at the 50th percentile of performance for unseen continuous-valued geometries.



(b) Output predictions at the 50th percentile of performance for unseen continuous-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. Predictions are able to consistently capture complex structures without graininess.

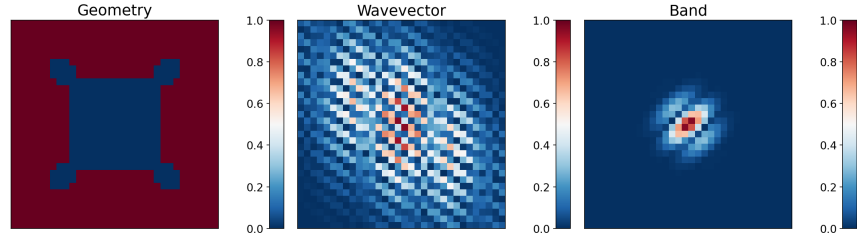


(a) Input sample at the 75th percentile of performance for unseen continuous-valued geometries.

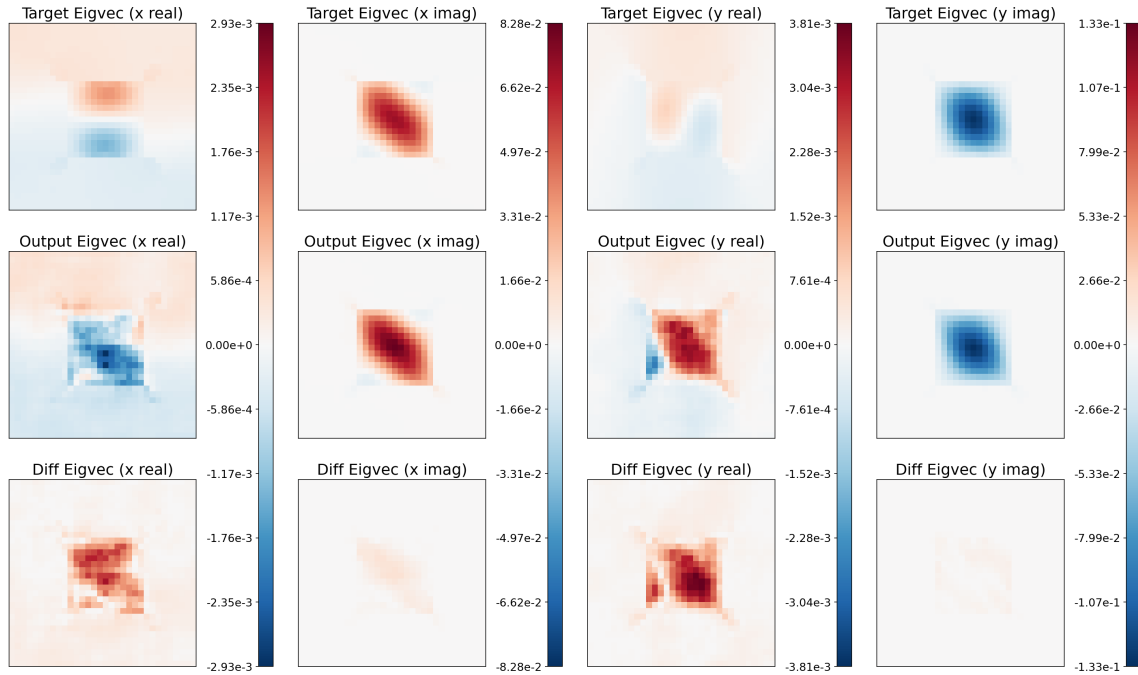


(b) Output predictions at the 75th percentile of performance for unseen continuous-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. At this point, the target and prediction are virtually indistinguishable by eye.

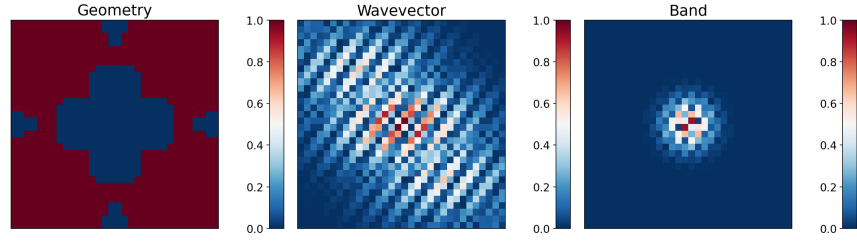
323 These results demonstrate that with wavelet encodings the FNO performs well
 324 on continuous geometries, faithfully reproducing the displacement fields in most
 325 cases. Model predictions in the upper performance quartiles match the targets well
 326 in structure and scale, with average pixel relative errors decreasing smoothly from
 327 on the order of 10^{-2} to 10^{-4} as performance percentile increases (see Figure 14a for
 328 detailed statistics).



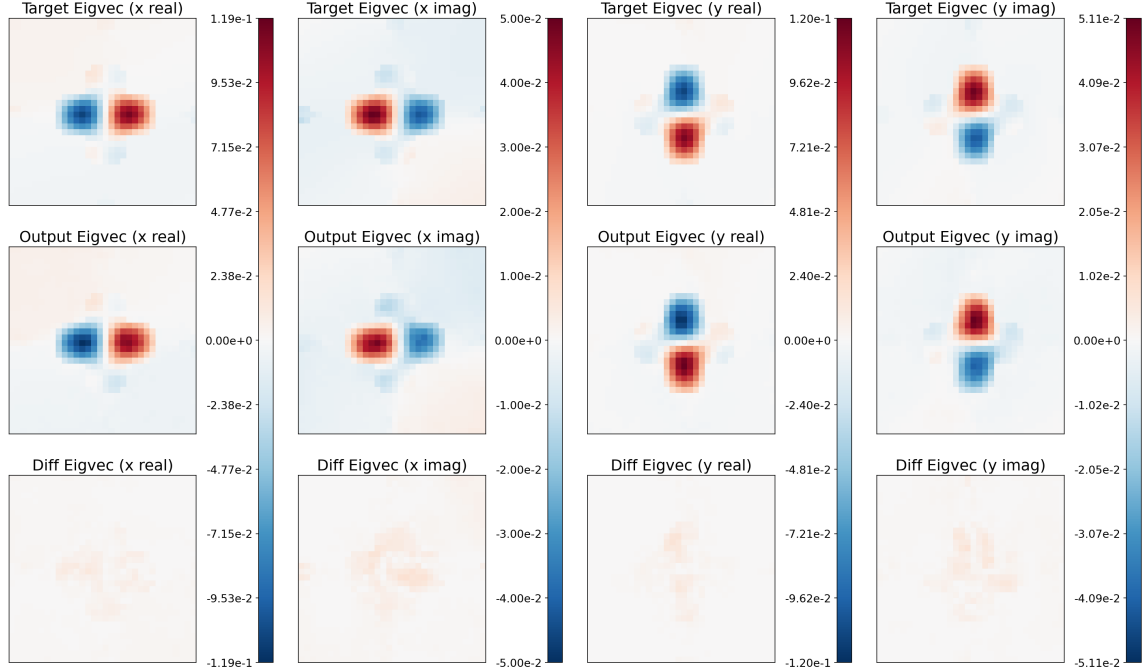
(a) Input sample at the 25th percentile of performance for unseen discrete-valued geometries.



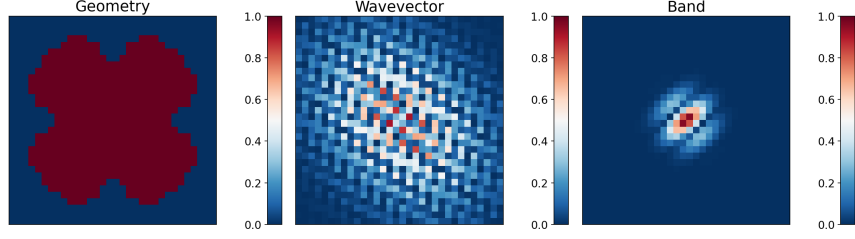
(b) Output predictions at the 25th percentile of performance for unseen discrete-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. Because NMAE normalizes residuals by target magnitude, low-amplitude channels receive greater relative weight during training than their absolute scale would suggest. For this sample the imaginary displacement channels are one to two orders of magnitude larger than the real channels, so absolute error can remain low overall while relative error stays large on the smaller-magnitude channels. (Note that the colorbars for each column are independently scaled.)



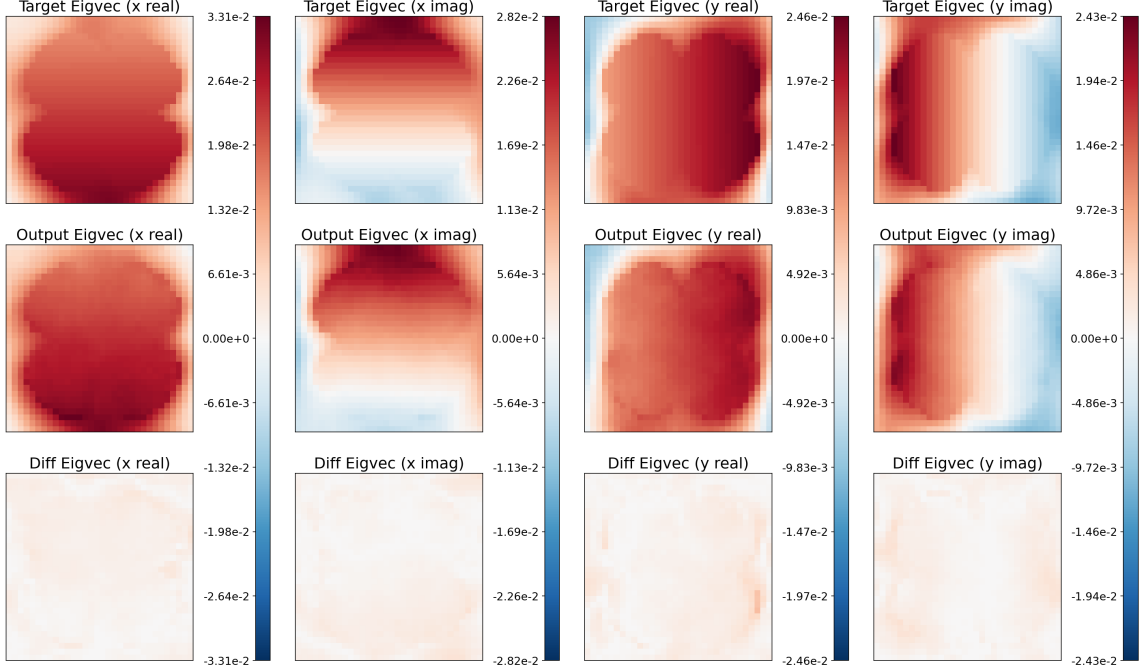
(a) Input sample at the 50th percentile of performance for unseen discrete-valued geometries.



(b) Output predictions at the 50th percentile of performance for unseen discrete-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. Absolute and relative errors are low, with strong agreement in structure and scale between targets and predictions.



(a) Input sample at the 75th percentile of performance for unseen discrete-valued geometries.



(b) Output predictions at the 75th percentile of performance for unseen discrete-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. At this point, disagreements between targets and predictions become very hard to see by eye.

330 As in the continuous-geometry case, these results demonstrate that with wavelet
 331 encodings the FNO performs well on discrete geometries, faithfully reproducing
 332 the displacement fields in most cases. Model predictions in the upper performance
 333 quartiles match the targets well in structure and scale, with average pixel relative
 334 errors decreasing smoothly from on the order of 10^{-2} to 10^{-4} as performance percentile
 335 increases (see Figure 14b for detailed statistics).

336 3.1.3. Continuous Versus Discrete Performance

337 That a single model predicts well on both continuous and discrete geometries is a
338 nontrivial result for FNOs paired with wavelet encodings. Prior encoding strategies
339 based on constant fields and sinusoidal fields performed far worse, producing grainy
340 structures or mode collapse, because conditioning information for the wavevector and
341 band could not propagate effectively through the spectral branch of the FNO layers.
342 Performance over all unseen samples in the test set is shown in Figure 14.

343 The poorer performance on binarized geometries is consistent with basic operator-
344 learning theory and known spectral limitations of Fourier-parameterized networks [24,
345 20, 25]. Continuous material fields lie in smoother function classes, whereas binary two-
346 phase designs contain jump discontinuities with nonexistent derivatives at material
347 interfaces. Since an FNO approximates the FEA operator $\mathcal{G} : \mathcal{U} \rightarrow \mathcal{V}$ through
348 truncated Fourier representations, it is naturally advantaged when both inputs and
349 outputs are comparatively smooth. Discrete geometries thus perform worse due to
350 sharp boundaries, localized gradients, and broadband high-wavenumber content that
351 are poorly captured on a fixed 32×32 grid and are mismatched to smooth wavelet
352 encodings. Additionally, continuous geometries more often contain large regions of
353 near-zero displacement, which inflate relative error through a small denominator.

354 These results indicate two important takeaways. The first is that FNOs with
355 wavelet encodings can learn multiple deformation modes for each geometry, which
356 the user can select by setting the wavevector and band inputs. The second is that
357 the same model can learn across continuous and discrete geometry distributions,
358 suggesting that some aspects of the underlying eigenvalue problem may be captured
359 in a geometry-agnostic way. Exploring that possibility more systematically is left for
360 future work.

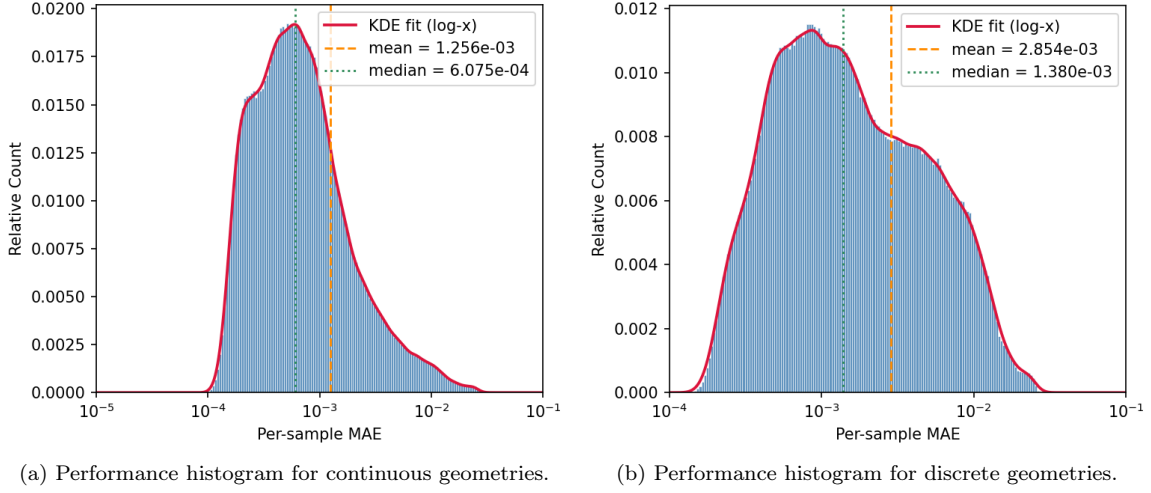


Figure 14: Comparison of relative prediction error distributions for continuous-valued and discrete metamaterial geometries for the same model. While there are some tail outliers, most samples fall within the range of 10^{-2} to 10^{-4} , indicating that the same model is able to learn continuous and discrete geometry cases.

3.2. Dispersion Band Reconstruction

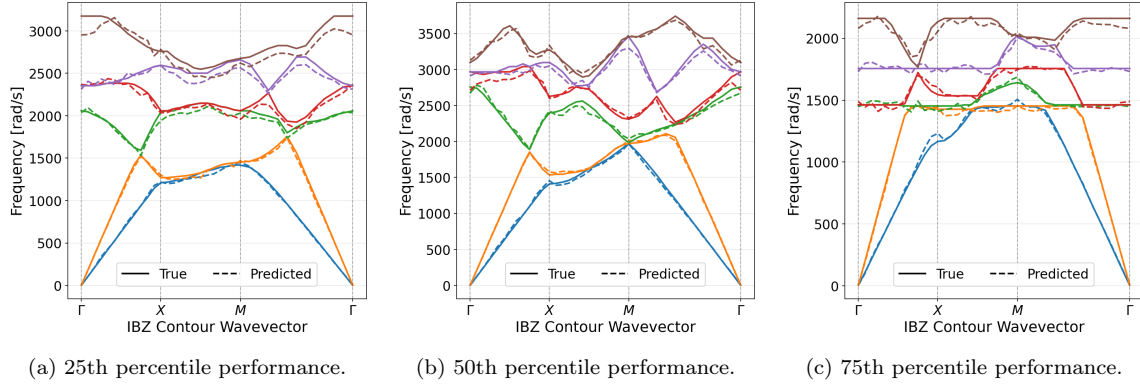


Figure 15: Predicted and ground-truth dispersion bands for representative unseen continuous-valued geometries spanning the 25th, 50th, and 75th percentiles of prediction performance.

These plots show eigenfrequency predictions across all 325 wavevectors of the IBZ half-plane for continuous-valued geometries, restricted to the horizontal, vertical, and diagonal IBZ traversals. Because the eigenfrequency is encoded with a logarithmic

transform to support multi-scale learning, errors tend to scale with the target magnitude, and higher bands can appear more deviant on a linear plot. Overall, the model achieves an average prediction error below 1% on the encoded eigenfrequency across unseen samples.

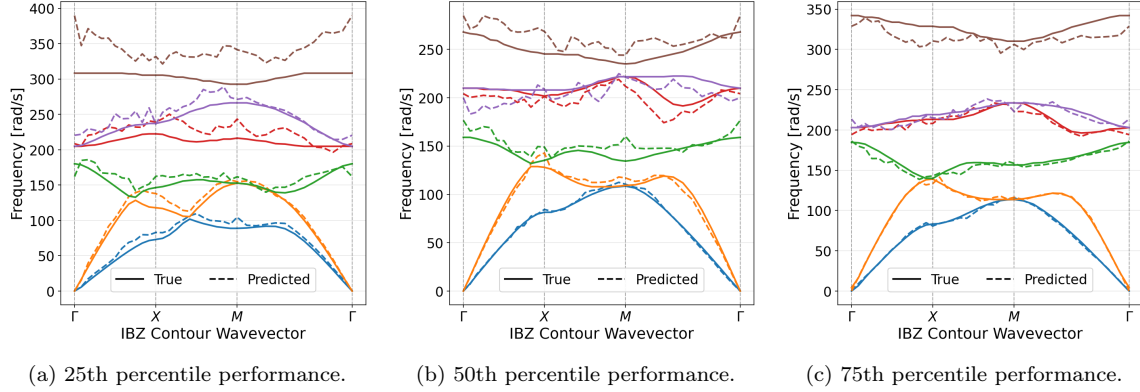


Figure 16: Predicted and ground-truth dispersion bands for representative unseen discrete-valued geometries spanning the 25th, 50th, and 75th percentiles of prediction performance.

As in the continuous-geometry figures, these plots show eigenfrequency predictions across all 325 wavevectors of the IBZ half-plane for discrete-valued geometries, restricted to the horizontal, vertical, and diagonal IBZ traversals. Overall, the model again achieves an average prediction error below 1% on the encoded eigenfrequency across unseen samples.

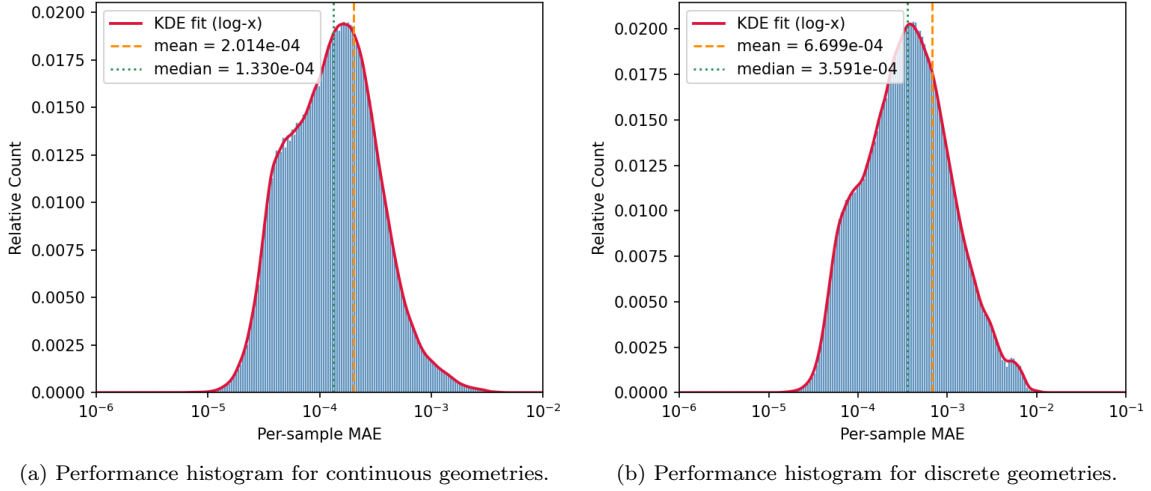


Figure 17: Comparison of relative prediction error distributions for continuous-valued and discrete metamaterial geometries for the same model. While there are some tail outliers, most samples fall within the range of 10^{-2} to 10^{-4} , indicating that the same model is able to learn continuous and discrete geometry cases.

3.3. Dependence of Prediction Error on Band and Boundary Length

The preceding subsections showed that, although the same wavelet-conditioned FNO can predict displacement fields for both continuous and discrete unit cells, discrete geometries are systematically more difficult, with absolute and relative errors larger (Figures 14, 17), and the harder quartiles of the discrete test set degrading more visibly than their continuous counterparts. This gap is expected from the architecture of the FNO itself and from the well-documented preference of spectral networks for low-frequency content [24, 20]. Each Fourier layer retains only a truncated set of spectral modes after the discrete FFT. Smooth continuous material fields, and the comparatively smooth displacement modes they induce, concentrate their energy in a modest number of low-to-moderate wavenumbers and are therefore well represented by that truncated Fourier basis. Binary geometries, by contrast, contain jump discontinuities at material interfaces. Under an FFT, such discontinuities produce a broadband spectrum whose coefficients decay slowly with wavenumber, so a substantial fraction of the high-frequency content needed to resolve sharp phase boundaries lies outside the retained modes; related analyses of Fourier-based solvers

390 for discontinuous coefficients likewise highlight Gibbs-type artifacts and degraded
 391 high-wavenumber recovery [25]. The spectral branch therefore sees a degraded, band-
 392 limited version of the interface, yielding Gibbs-type ringing or blur and a harder
 393 operator-learning problem precisely when geometric discontinuity is more severe. If
 394 this spectral-truncation explanation is correct, then even among discrete geometries
 395 the prediction error should increase with the boundary interface length. To test this
 396 empirically, we define the *boundary length* of each binarized unit cell as the number of
 397 interior four-connected pixel edges separating a 0-valued pixel from a 1-valued pixel.
 398 This scalar is a discrete measure of interface complexity. For each of the 1000 unseen
 399 discrete test geometries, we compute the mean absolute error (MAE) of the predicted
 400 displacement channels, averaged over all 325 IBZ wavevectors, separately for each of
 401 the six retained eigenbands. The resulting relationships are shown in Figure 18.

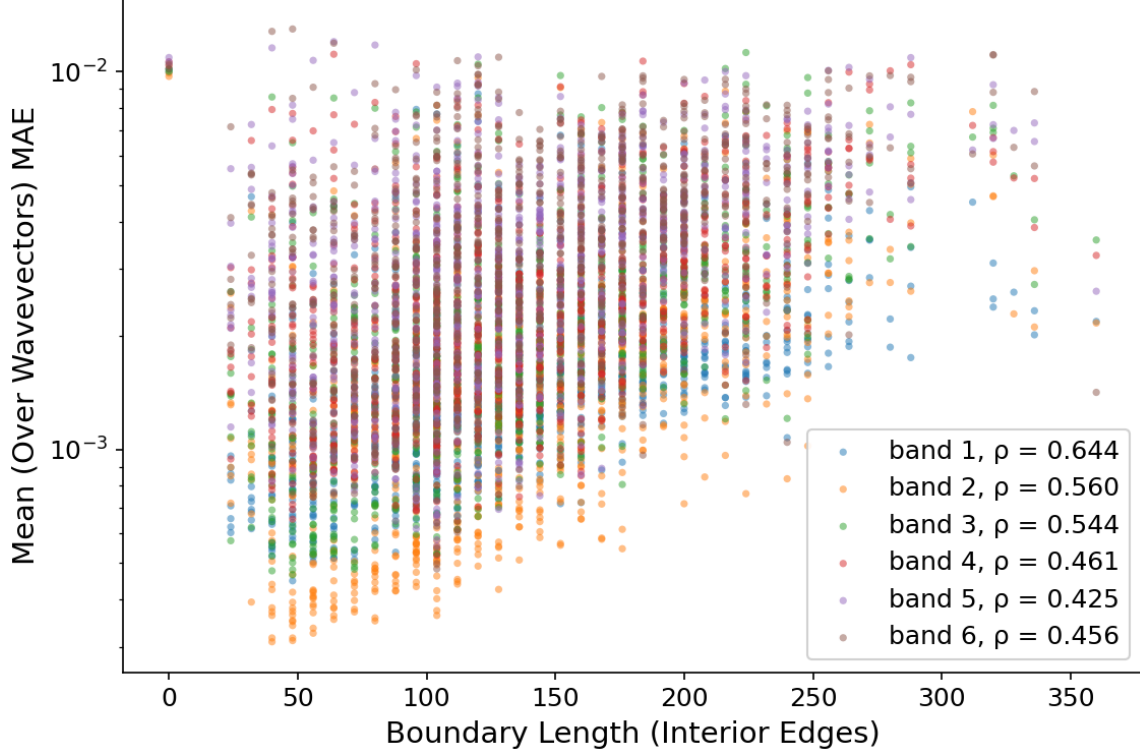


Figure 18: Mean displacement MAE versus interface boundary length, separated by color for each of the six eigenbands on the discrete test set. Points correspond to average performance on unit-cell geometries. Legend values ρ are Spearman rank correlations. A positive trend appears for every band, consistent with degraded FNO accuracy as the total length of 0/1 material interfaces increases.

Two complementary trends are apparent. First, for every band there is a positive monotonic association between boundary length and MAE, with Spearman correlations ranging from $\rho = 0.644$ for band 1 to $\rho \approx 0.42$ – 0.56 for the higher bands. Geometries with short interfaces are concentrated at lower absolute error, while geometries with long, meandering phase boundaries systematically occupy the upper portion of the cloud. This supports the interpretation that the hard cases for the learned operator are those for which the truncated Fourier representation must resolve more extensive sharp material transitions on the fixed 32×32 grid. Second, absolute error itself increases with band index. Averaged over the test geometries, the mean wavevector-averaged MAE rises from approximately 1.6×10^{-3} for band 1 to approximately 3.8×10^{-3} for band 6. Higher bands correspond to more rapidly oscillating

413 Bloch modes that concentrate energy nearer material interfaces and therefore inherit
 414 more of the geometric difficulty encoded by boundary length. Together with the
 415 continuous-versus-discrete comparisons above, Figure 18 indicates that prediction
 416 difficulty on discrete designs is organized both by band (modal complexity) and by
 417 interface measure (geometric complexity), as expected when a spectral surrogate
 418 retains only a portion of the broadband Fourier content generated by discontinuities.

419 3.4. Band and Wavevector Encoding

420 For the FNO to ingest the conditioning constants of the eigenvalue PDE in equation
 421 1, namely the wavevectors k_x , k_y and the band index b that selects which eigenmode
 422 to return, these scalars must be represented as matrices or tensors compatible with
 423 the existing channel wise input structure. Alternatives that inject scalars through
 424 specialized pathways would require heavy architectural alterations, which are hard
 425 to justify given that FNO theory suggests eigenvalue PDEs should be solvable with
 426 existing hyperparameter combinations. We therefore compared three strategies for
 427 encoding these constants as input fields: constant valued fields, spatial sinusoids, and
 428 the Gabor wavelet encoding described in Section 2.4.

429 The constant field representation is the naive default. On pixel coordinates
 430 $(x, y) \in \{0, \dots, S - 1\}$ with $S = 32$, wavevector and band were encoded as

$$I_{k_x}(x, y) = \frac{k_x}{\pi}, \quad I_{k_y}(x, y) = \frac{k_y}{\pi}, \quad k_x, k_y \in [-\pi, \pi],$$

431

$$I_b(x, y) = \frac{b}{10}, \quad b \in \{1, 2, 3, 4, 5, 6\}.$$

432 Each field is spatially uniform: the normalized scalars $\frac{k_x}{\pi}, \frac{k_y}{\pi} \in [-1, 1]$ and $\frac{b}{10} \in$
 433 $[0.1, 0.6]$ are broadcast over the $S \times S$ grid. Because the FFT of a constant field is
 434 zero everywhere except at the DC coefficient, the frequency-domain branch of each
 435 Fourier layer receives little non-trivial information to propagate beyond the first layer,
 436 so learning must rely mainly on the shallow spatial-domain branch. In practice this
 437 encoding is fragile: under some training hyperparameters, particularly high learning
 438 rates, training collapses to near-constant outputs that ignore the conditioning inputs.

439 With more stable settings the model can still learn, but it underperforms the wavelet
 440 encoding. The best uniform-encoding results we obtained under matched conditions
 441 are reported in Tables 2 and 3 below.

442 A spatial sinusoidal encoding restores unique and nondegenerate representations in
 443 the spectral domain and allows the FNO to learn the problem. On pixel coordinates
 444 $(x, y) \in \{0, \dots, S - 1\}$, wavevector and band were encoded as

$$I_k(x, y) = \sin\left(\frac{2}{S}(k_x x + k_y y)\right), \quad k_x, k_y \in [-\pi, \pi].$$

445

$$I_b(x, y) = \frac{1}{2} \left[\cos\left(\frac{2\pi b x}{S}\right) + \cos\left(\frac{2\pi b y}{S}\right) \right], \quad b \in \{1, 2, 3, 4, 5, 6\}.$$

446 The wavevector field is a directed plane wave: its phase gradient is parallel to (k_x, k_y) ,
 447 so wavefronts propagate along the physical wavevector, and the spatial frequency
 448 scales with $|\mathbf{k}|$. The prefactor $2/S$ places about one cycle across the patch at the
 449 IBZ boundary $|\mathbf{k}| = \pi$, which keeps neighboring (k_x, k_y) pairs visually and spectrally
 450 distinct on the $S \times S$ grid while avoiding severe aliasing. A sine rather than a cosine is
 451 used so that the Γ point ($\mathbf{k} = \mathbf{0}$) maps to the zero field instead of a constant DC patch.
 452 However, because a sinusoid is spectrally sparse, with energy concentrated at only a
 453 few wavenumbers, the Fourier branch remains weakly utilized, and predictions exhibit
 454 a persistent grainy, under-resolved structure relative to ground truth. Increasing
 455 model depth and width did not remove this graininess, indicating that the bottleneck
 456 was the encoding rather than model capacity.

457 The Gabor wavelet encoding resolves this by design. Following Gabor’s classical
 458 construction [21], wavelets are constructed to trade certainty between the spatial
 459 and spectral domains: neither representation is infinitely sharp, but both remain
 460 informationally rich. As a result, the conditioning inputs populate both branches of
 461 each Fourier layer with usable structure, both branches can train and contribute, and
 462 the model yields the accurate, well resolved predictions reported above while enabling
 463 deterministic mode selection. These observations support the broader conclusion
 464 that input encodings should be matched to the spatial and spectral structure of the
 465 operator: encodings that populate only one domain underuse the FNO’s Fourier

branch, whereas wavelet encodings exercise both.

To quantify these differences under matched training conditions, we compare wavelet-, sinusoidal-, and uniform-encoded models on held-out continuous and discrete test geometries using overall-sample MAE, MSE, NMAE, and NMSE (Tables 2 and 3). All three runs use the same training hyperparameters: an FNO with 4 Fourier layers and 128 hidden channels, AdamW optimization, learning rate 2×10^{-3} with StepLR step size 1 and per-epoch decay 0.9, batch size 520, and an NMAE training objective. Checkpoints are taken after the 8th training epoch, where the sinusoidal- and uniform-encoded models begin to asymptote. The wavelet-encoded model continues to improve until roughly epoch 12, but the epoch-8 checkpoint is reported here for a fair comparison at equal training budget. Positive percentages in the tables indicate worse error than the wavelet-encoded model.

Geometry Dataset	Median Loss	Wavelet Encoding	Sinusoidal Encoding	Versus Wavelet	Uniform Encoding	Versus Wavelet
Continuous	MAE	5.621×10^{-4}	7.393×10^{-4}	+31.6%	7.156×10^{-4}	+27.3%
	MSE	1.236×10^{-6}	1.825×10^{-6}	+47.7%	1.848×10^{-6}	+49.5%
	NMAE	9.13×10^{-2}	1.202×10^{-1}	+31.6%	1.124×10^{-1}	+23.1%
	NMSE	7.971×10^{-3}	1.135×10^{-2}	+42.4%	1.169×10^{-2}	+46.7%
Discrete	MAE	1.336×10^{-3}	1.488×10^{-3}	+11.4%	1.520×10^{-3}	+13.8%
	MSE	5.004×10^{-6}	5.712×10^{-6}	+14.2%	6.150×10^{-6}	+22.9%
	NMAE	1.800×10^{-1}	1.996×10^{-1}	+10.9%	2.051×10^{-1}	+13.9%
	NMSE	3.01×10^{-2}	3.38×10^{-2}	+12.2%	3.76×10^{-2}	+24.7%

Table 2: Overall-sample median test losses and relative change versus the wavelet-encoded model. Lower is better.

Geometry Dataset	Mean Loss	Wavelet Encoding	Sinusoidal Encoding	Versus Wavelet	Uniform Encoding	Versus Wavelet
Continuous	MAE	1.405×10^{-3}	1.649×10^{-3}	+17.4%	1.591×10^{-3}	+13.2%
	MSE	3.234×10^{-5}	3.496×10^{-5}	+8.1%	3.529×10^{-5}	+9.1%
	NMAE	2.876×10^{-1}	3.559×10^{-1}	+23.7%	3.466×10^{-1}	+20.5%
	NMSE	1.405×10^{-1}	1.615×10^{-1}	+14.9%	1.619×10^{-1}	+15.3%
Discrete	MAE	2.662×10^{-3}	2.830×10^{-3}	+6.3%	2.861×10^{-3}	+7.5%
	MSE	6.641×10^{-5}	6.756×10^{-5}	+1.7%	7.030×10^{-5}	+5.8%
	NMAE	4.905×10^{-1}	5.416×10^{-1}	+10.4%	5.585×10^{-1}	+13.9%
	NMSE	3.435×10^{-1}	3.500×10^{-1}	+1.9%	3.776×10^{-1}	+9.9%

Table 3: Overall-sample mean test losses and relative change versus the wavelet-encoded model. Lower is better.

4. Conclusions

This work shows that a single Fourier Neural Operator can learn multiple eigenmodes of the elastic wave equation for acoustic metamaterials, recovering Bloch displacement fields and eigenfrequencies across both continuous and discrete unit cell geometries. Relative to prior surrogates that mainly target dispersion curves, transmission spectra, or bandgap scalars [12, 13, 14], the present model returns full deformation modes with deterministic selection among modes through wavelet encodings of the wavevector and band index. On unseen geometries of both types, displacement predictions typically achieve relative errors between 10^{-2} and 10^{-4} , and reconstructed dispersion relations average below 1% error. Encoding choice is essential for this capability. Constant field encodings are spectrally fragile. Sinusoidal encodings train but remain grainy. Gabor wavelet encodings [21] supply usable structure in both the spatial and spectral branches of the FNO, enabling reliable mode selection.

The same spectral perspective explains the observed limits. Prediction error grows with interface length in discrete geometries and with band index, consistent with truncated Fourier representations struggling to resolve sharp material boundaries and rapidly oscillating modes [24, 20, 25]. Even so, the trained surrogate is lightweight (about 2 GB at float16) and evaluates in about 1 ms per sample on a consumer-grade

497 CPU, compared with about 1 s per sample for the corresponding FEA eigenvalue
498 solve. That three orders of magnitude speedup makes repeated Bloch analyses, which
499 are central to optimization, uncertainty quantification, inverse design, and screening
500 of unit cell geometries, far more practical than direct finite element evaluation at
501 every design iterate. In short, FNOs with wavelet encodings of modal parameters
502 provide a pathway to greatly accelerate acoustic metamaterial simulation and design
503 relative to repeated finite element eigenvalue analysis. Future work includes other
504 spatial resolutions, physics informed operator losses [9], and broader design spaces
505 beyond the present eight fold symmetric, two material unit cells, including other
506 tiling symmetries, asymmetric geometries, and multimaterial compositions.

507 **AI Usage Disclosure**

508 Generative AI tools were used to assist with coding of experiments, translating
509 MATLAB code to Python, and automating training pipelines. AI tools were also used
510 for light editing of the manuscript. All scientific claims, results, and final wording
511 were reviewed and approved by the authors.

512 **Declaration of Competing Interest**

513 The authors declare that they have no known competing financial interests or
514 personal relationships that could have appeared to influence the work reported in
515 this paper.

516 **CRedit authorship contribution statement**

517 **Han Zhang:** Conceptualization, Data curation, Formal analysis, Investigation,
518 Methodology, Software, Validation, Visualization, Writing – original draft, Writing –
519 review and editing.

520 **Alexander Ogren:** Data curation, Software.

521 **Cynthia Rudin:** Supervision, Writing – review and editing.

522 **Johann Guilleminot:** Formal analysis, Methodology, Supervision, Writing – review
523 and editing.

524 **L. Catherine Brinson:** Conceptualization, Funding acquisition, Methodology,
525 Project administration, Resources, Software, Supervision, Visualization, Writing –
526 review and editing.

527 **Data availability**

528 Python code and experiments can be found at [https://github.com/trutheresy/](https://github.com/trutheresy/NO-2D-Metamaterials)
529 [NO-2D-Metamaterials](https://github.com/trutheresy/NO-2D-Metamaterials).

530 MATLAB FEA code can be found at <https://github.com/aco8ogren/2D-dispersion>.
531 [git](https://github.com/aco8ogren/2D-dispersion).

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613 S1. Supplementary Information

614 S1.1. Model and Training Hyperparameters

Hyperparameter	Candidate Settings			
Fourier Layers	4	6	8	12
Hidden Channels	32	64	128	256
Activation Function	ReLU	GELU	tanh	Sigmoid
Loss Criterion	L1	L2	NMAE	SSIM
Epochs	8	12	16	20
Learning Rate	10^{-4}	2×10^{-3}	10^{-2}	10^{-1}
Weight Decay	0	10^{-4}	10^{-3}	10^{-2}
Gamma	0	0.01	0.1	0.5
Step Size	0	1	4	10
Batch Size	64	128	256	520
Data Precision	F8 E4M3	F8 E5M2	F16	F32

Table S1: Combinations tried in grid search for optimal model and training hyperparameters.

615 S1.2. Loss Criterion

616 In the exploration of optimal training protocols for the FNO model, several loss
617 functions were evaluated as training objectives: mean absolute error (MAE), mean
618 squared error (MSE), normalized MAE (NMAE), normalized MSE (NMSE), and
619 structural similarity index (SSIM). Among these, NMAE yielded the best overall
620 performance and was used for all reported results. All comparisons used a learning
621 rate of 2×10^{-3} with a per-epoch decay factor of 0.9. The mathematical definitions of
622 each loss follow. Losses were evaluated and aggregated across all five output channels.

- 623 • The mean absolute error (MAE) loss is defined as

$$\mathcal{L}_{\text{MAE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 |\hat{u}_c(i, j) - u_c(i, j)|$$

624 Experimentally, models trained exclusively with MAE produced predictions
625 with sharper interfaces and better-preserved boundary features than those

626 trained solely with MSE. This behavior is consistent with the linear penalty
 627 of MAE, which does not disproportionately penalize isolated large residuals
 628 near discontinuities. Despite the improved qualitative sharpness, MAE training
 629 underperformed NMAE on overall validation accuracy, so absolute-error training
 630 alone was not selected for the reported models.

- 631 • The mean squared error (MSE) loss is defined as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 (\hat{u}_c(i, j) - u_c(i, j))^2$$

632 The quadratic penalty of MSE assigns increasingly large gradients to larger
 633 residuals, causing the optimizer to preferentially reduce regions with the greatest
 634 numerical error. As a consequence, the loss favors smooth predictions that min-
 635 imize the overall energy of the residual rather than preserving sharp transitions.
 636 Experimentally, models trained exclusively with MSE produced outputs that
 637 reproduced the correct magnitude and large-scale structure of the solution, but
 638 exhibited diffuse interfaces and blurred boundaries. Fine-scale features were
 639 often replaced by low-amplitude, grainy artifacts, consistent with the tendency
 640 of MSE to average over uncertain or discontinuous regions in order to minimize
 641 the global squared error.

- 642 • The structural similarity index measure (SSIM) is defined as

$$\mathcal{L}_{\text{SSIM}} = 1 - \frac{1}{5} \sum_{c=0}^4 \frac{(2\mu_{u_c}\mu_{\hat{u}_c} + C_1)(2\sigma_{u_c\hat{u}_c} + C_2)}{(\mu_{u_c}^2 + \mu_{\hat{u}_c}^2 + C_1)(\sigma_{u_c}^2 + \sigma_{\hat{u}_c}^2 + C_2)}, \quad \text{where}$$

$$\mu_{u_c} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} u_c(i, j), \quad \sigma_{u_c\hat{u}_c} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} (u_c(i, j) - \mu_{u_c})(\hat{u}_c(i, j) - \mu_{\hat{u}_c})$$

644 A significant limitation of SSIM arises when it is applied to signed-valued
 645 fields. Unlike its original application to nonnegative image intensities, a perfect
 646 sign-reversed field ($\hat{u} = -u$) can achieve nearly the same SSIM score as a

perfect reconstruction ($\hat{u} = u$). Consequently, a model trained solely with SSIM may converge to physically incorrect solutions containing polarity inversions while still minimizing the loss. The following derivation demonstrates the mathematical origin of this ambiguity.

For the purposes of the following discussion, we consider the common case where the stabilizing constants satisfy

$$C_1 \ll 2\mu_{u_c}^2, \quad C_2 \ll 2\sigma_{u_c}^2,$$

allowing them to be neglected.

For a correct reconstruction,

$$u_c^{(+)} = u_c,$$

the SSIM score is

$$\text{SSIM}(u_c, u_c^{(+)}) = \frac{(2\mu_{u_c}^2)(2\sigma_{u_c}^2)}{(2\mu_{u_c}^2)(2\sigma_{u_c}^2)} = 1.$$

Now consider a sign-reversed reconstruction. The field and its mean reverse sign,

$$u_c^{(-)} = -u_c, \quad \mu_{u_c^{(-)}} = -\mu_{u_c},$$

while the covariance reverses sign and the variance remains unchanged,

$$\sigma_{u_c, u_c^{(-)}} = -\sigma_{u_c, u_c}, \quad \sigma_{u_c^{(-)}}^2 = \sigma_{u_c}^2.$$

Substituting these relationships into the SSIM expression gives

$$\text{SSIM}(u_c, u_c^{(-)}) = \frac{(-2\mu_{u_c}^2)(-2\sigma_{u_c}^2)}{(2\mu_{u_c}^2)(2\sigma_{u_c}^2)} = 1.$$

The numerator therefore differs from that of a correct reconstruction by two factors of -1 , and thus, under these conditions, SSIM assigns essentially the same similarity score to a field and its sign-reversed counterpart as it does to

a perfect reconstruction. This ambiguity constitutes a significant limitation when SSIM is applied to signed-valued regression problems where the polarity of the solution is physically meaningful. During training, the network frequently converged to predictions in which localized regions or the entire image were the negative of the correct solution while still achieving a favorable SSIM objective. Although the resulting fields exhibited accurate interface locations, boundary geometry, and overall morphology, the polarity inversions rendered the predictions physically incorrect. Consequently, SSIM should not be used as the sole training criterion for signed-valued fields unless it is supplemented by an additional loss that explicitly penalizes sign reversals.

- The normalized mean absolute error (NMAE) loss is defined as

$$\mathcal{L}_{\text{NMAE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 \frac{|\hat{u}_c(i, j) - u_c(i, j)|}{|u_c(i, j)| + \varepsilon}$$

Unlike MAE, NMAE weights each prediction error by the inverse magnitude of the target value, causing the optimization to place greater emphasis on regions where the ground-truth solution is small in scale. Consequently, the loss seeks to minimize relative error rather than absolute error, preventing low-magnitude features from being dominated by high-magnitude regions during training.

Experimentally, NMAE gave the best overall balance across channels with disparate magnitudes and was selected as the training objective for all reported results. Relative-error emphasis improved reconstruction of low-amplitude displacement components without sacrificing usable accuracy on larger-magnitude channels, which made NMAE preferable to MAE, MSE, NMSE, and SSIM for this application.

- The normalized mean squared error (NMSE) loss is defined as

$$\mathcal{L}_{\text{NMSE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 \frac{(\hat{u}_c(i, j) - u_c(i, j))^2}{(u_c(i, j))^2 + \varepsilon}$$

686 Similar to NMAE, NMSE normalizes the prediction error by the magnitude
687 of the target field, causing the optimization to prioritize relative error rather
688 than absolute error. The quadratic penalty retains the characteristic behavior
689 of MSE by assigning greater weight to larger residuals, while the normalization
690 emphasizes regions where the target magnitude is small.

691 Experimentally, NMSE reduced the relative error in low-amplitude regions com-
692 pared with standard MSE, but it underperformed NMAE on overall validation
693 accuracy and was not selected for the reported models.

694 *S1.3. Algorithms for Input Wavelet Encoding*

695 The band and wavevector conditioning channels used by the FNO are generated
696 by the Gabor wavelet embeddings below, which match the implementations in the
697 accompanying code.

Algorithm 1 1D Gabor Wavelet Embedding from Band Index

- 1: **INPUT:** Band index b ; grid size $S \leftarrow 32$; frequency scale $r \leftarrow 2.0$
- 2: Create linspace vectors $x, y \in [-1, 1]$ with S points
- 3: Construct meshgrid $X, Y \leftarrow \text{meshgrid}(x, y)$
- 4: Compute base frequency:

$$f \leftarrow (1.0 + |b|) \cdot \frac{S}{8}$$

- 5: Compute rotation angle:

$$\theta \leftarrow (b \bmod 8) \cdot \frac{S}{16}$$

- 6: Rotate coordinates:

$$X_\theta \leftarrow X \cos \theta + Y \sin \theta, \quad Y_\theta \leftarrow -X \sin \theta + Y \cos \theta$$

- 7: Set Gaussian envelope widths:

$$\sigma_x \leftarrow \sigma_y \leftarrow \frac{0.3}{r}$$

- 8: Compute Gabor wavelet embedding:

$$\psi_b(x, y) \leftarrow \exp \left(-\frac{X_\theta^2}{2\sigma_x^2} - \frac{Y_\theta^2}{2\sigma_y^2} \right) \cdot \cos(f \cdot X_\theta)$$

- 9: **RETURN:** $\psi_b \in \mathbb{R}^{S \times S}$
-

698 In Algorithm 1, the grid size $S = 32$ matches the geometry resolution used by
699 the FNO. For this resolution the base frequency is $f = (1 + |b|) S/8$, so successive
700 bands increase both carrier frequency and orientation. The modulo in θ repeats
701 orientation every eight bands, while f continues to increase, so embeddings remain
702 distinguishable beyond eight bands. The constants $r = 2.0$, $S/16$, and 0.3 were chosen
703 so that the wavelets occupy most of the spatial domain and remain visually distinct
704 across bands.

Algorithm 2 2D Gabor Wavelet Embedding for Wavevectors

- 1: **INPUT:** Wavevector components k_x, k_y (radians); grid size $S \leftarrow 32$; frequency scale $r \leftarrow 1.0$
- 2: Set constants $f_0 \leftarrow 2.20$, $\alpha \leftarrow 41$, $N_x \leftarrow 13$, $N_y \leftarrow 25$
- 3: Create linspace vectors $x, y \in [-1, 1]$ with S points
- 4: Construct meshgrid $X, Y \leftarrow \text{meshgrid}(x, y)$
- 5: Compute frequencies:

$$f_x \leftarrow (f_0 + k_x) \alpha, \quad f_y \leftarrow (f_0 + k_y) \alpha$$

- 6: Compute rotation angles:

$$\theta_x \leftarrow (k_x \bmod N_x) \cdot \frac{\pi}{N_x}, \quad \theta_y \leftarrow (k_y \bmod N_y) \cdot \frac{\pi}{N_y}$$

- 7: Rotate coordinates:

$$X_\theta \leftarrow X \cos \theta_x + Y \sin \theta_y, \quad Y_\theta \leftarrow -X \sin \theta_x + Y \cos \theta_y$$

- 8: Set Gaussian envelope widths:

$$\sigma_x \leftarrow \sigma_y \leftarrow \frac{0.5}{r}$$

- 9: Compute Gabor wavelet embedding:

$$\psi_{k_x, k_y}(x, y) \leftarrow \exp \left(-\frac{X_\theta^2}{2\sigma_x^2} - \frac{Y_\theta^2}{2\sigma_y^2} \right) \cdot \sin(f_x X_\theta) \cdot \sin(f_y Y_\theta)$$

- 10: **RETURN:** $\psi_{k_x, k_y} \in \mathbb{R}^{S \times S}$
-

705 In Algorithm 2, k_x and k_y enter as continuous radian valued wavevector com-
706 ponents. Distinct modulo periods $N_x = 13$ and $N_y = 25$ reduce aliasing under
707 interchange of the two components. The offset f_0 and scale α map the physical
708 range of k into carrier frequencies that remain well resolved on the $S \times S$ grid, while

709 $\sigma = 0.5/r$ sets the envelope width so that the patterns remain spatially localized yet
710 informative in both domains.

711 *S1.4. Wavelet Decoding Fidelity*

712 For purposes of checking general applicability, an experiment was run to check
713 if a positive scalar may be embedded in a Gabor wavelet image and recovered
714 after pixelization, storage, or inference. Algorithms 3 and 4 implement this encode–
715 decode pair by spectral peak detection with centroid refinement. In the experiment,
716 we test recovery over $[1, 8000]$, spanning about four orders of magnitude. Despite
717 discretization onto a finite $S \times S$ grid and limited numeric precision associated with
718 float16 storage and arithmetic, the decoded values remain accurate across the full
719 range. Figures S1 and S2 show representative encodings and the corresponding round
720 trip error.

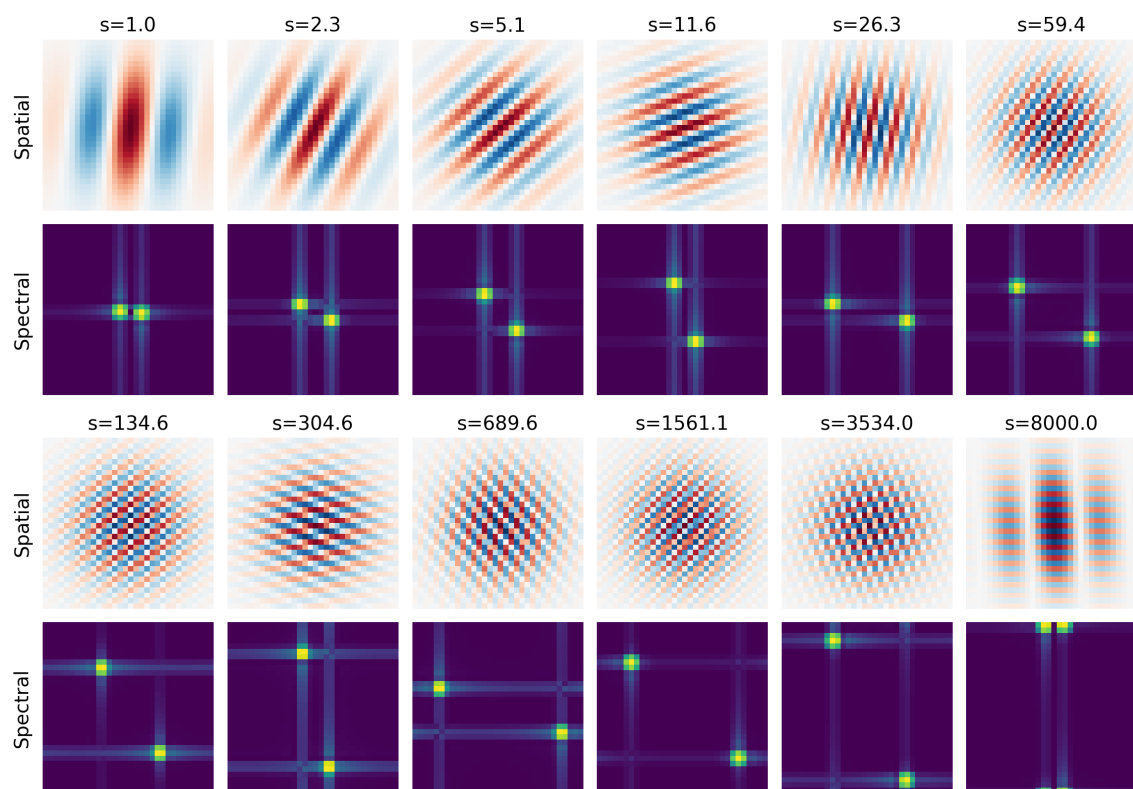


Figure S1: Log spaced Gabor wavelet encodings of scalars from 1 to 8000. Alternating rows show the spatial wavelet images and the corresponding Fourier magnitude spectra.

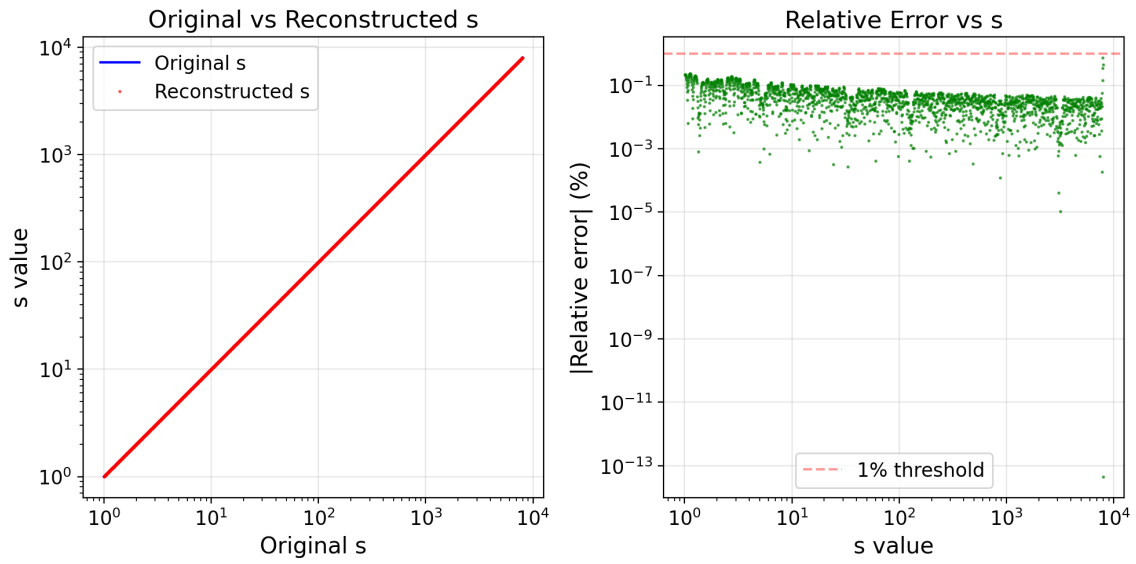


Figure S2: Encode–decode error for scalars from a minimum of 1 to a maximum of 8000.

Algorithm 3 Scalar Wavelet Encoding

- 1: **INPUT:** Scalar $s > 0$; image size $S \leftarrow 32$; bounds $[s_{\min}, s_{\max}]$; spectral-index bounds $[k_{\min}, k_{\max}]$
- 2: Set constants $\gamma \leftarrow 1$, $\phi \leftarrow 0$, $\sigma \leftarrow 8$, $\theta_{\min} \leftarrow \pi/30$, $\theta_{\max} \leftarrow \pi/2 - \pi/30$
- 3: Compute log-domain values:

$$\ell_s \leftarrow \log s, \quad \ell_{\min} \leftarrow \log s_{\min}, \quad \ell_{\max} \leftarrow \log s_{\max}$$

- 4: Compute normalized position and spectral index:

$$t \leftarrow \text{clip}\left(\frac{\ell_s - \ell_{\min}}{\ell_{\max} - \ell_{\min}}, 0, 1\right), \quad k \leftarrow k_{\min} + t(k_{\max} - k_{\min})$$

- 5: Compute log width per unit k :

$$\Delta_\ell \leftarrow \frac{\ell_{\max} - \ell_{\min}}{k_{\max} - k_{\min}}$$

- 6: Compute orientation from within-band log position:

$$\theta \leftarrow \theta_{\min} + \left(\frac{(\ell_s - \ell_{\min}) \bmod \Delta_\ell}{\Delta_\ell}\right) (\theta_{\max} - \theta_{\min})$$

- 7: Build centered grid $X, Y \in \{-S/2, \dots, S/2 - 1\}$, then rotate:

$$X_\theta \leftarrow X \cos \theta + Y \sin \theta, \quad Y_\theta \leftarrow -X \sin \theta + Y \cos \theta$$

- 8: Set envelope widths $\sigma_x \leftarrow \sigma$, $\sigma_y \leftarrow \gamma \sigma_x$, and carrier frequency $\omega \leftarrow 2\pi k/S$
- 9: Compute wavelet image:

$$\psi_s \leftarrow \exp\left(-\frac{1}{2} \left(\frac{X_\theta^2}{\sigma_x^2} + \frac{Y_\theta^2}{\sigma_y^2}\right)\right) \cos(\omega X_\theta + \phi)$$

- 10: **RETURN:** $\psi_s \in \mathbb{R}^{S \times S}$, k , θ
-

Algorithm 4 Scalar Wavelet Decoding

- 1: **INPUT:** Wavelet image $\psi \in \mathbb{R}^{S \times S}$; bounds $[s_{\min}, s_{\max}]$, $[k_{\min}, k_{\max}]$, $[\theta_{\min}, \theta_{\max}]$
- 2: Set constants for centroid refinement: radius $R \leftarrow 3$, power $p \leftarrow 2$
- 3: Mean-center image $\psi_c \leftarrow \psi - \text{mean}(\psi)$, then compute the centered 2D discrete Fourier magnitude:

$$\hat{\psi}(k_x, k_y) \leftarrow \sum_{m=0}^{S-1} \sum_{n=0}^{S-1} \psi_c(m, n) \exp\left(-2\pi i \left(\frac{k_x m}{S} + \frac{k_y n}{S}\right)\right), \quad F(k_x, k_y) \leftarrow |\hat{\psi}(k_x, k_y)|$$

- 4: Keep upper-half-plane spectral support (plus positive-frequency center row) and find dominant peak index (r^*, c^*)
- 5: Define symmetric peak partner about center, then compute weighted local centroid around (r^*, c^*) within radius R , using weights $w = F^p$ and excluding points closer to the symmetric partner
- 6: Convert refined centroid to wavevector coordinates (k_x, k_y) , then:

$$k_{\text{ext}} \leftarrow \sqrt{k_x^2 + k_y^2}, \quad \theta_{\text{ext}} \leftarrow \text{atan2}(k_y, k_x) \bmod \pi$$

- 7: Map extracted k to approximate log value:

$$\ell_{\text{approx}} \leftarrow \ell_{\min} + \text{clip}\left(\frac{k_{\text{ext}} - k_{\min}}{k_{\max} - k_{\min}}, 0, 1\right) (\ell_{\max} - \ell_{\min})$$

- 8: Let $\Delta_\ell \leftarrow (\ell_{\max} - \ell_{\min}) / (k_{\max} - k_{\min})$, compute within-band position from angle:

$$\alpha \leftarrow \text{clip}\left(\frac{\theta_{\text{ext}} - \theta_{\min}}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \quad \delta_\ell \leftarrow \alpha \Delta_\ell$$

- 9: Find nearest consistent log value by testing neighboring bands $b-1, b, b+1$:

$$\ell^* \leftarrow \arg \min_{\ell \in \{\ell_{\min} + j\Delta_\ell + \delta_\ell\}} |\ell - \ell_{\text{approx}}|$$

- 10: Clip ℓ^* to $[\ell_{\min}, \ell_{\max}]$, then decode:

$$s_{\text{ext}} \leftarrow \exp(\ell^*)$$

- 11: **RETURN:** $s_{\text{ext}}, k_{\text{ext}}, \theta_{\text{ext}}$
-

721 Algorithms 3 and 4 summarize the forward and inverse mapping used in this
722 experiment. The forward mapping places the scalar on a log scale and jointly encodes
723 magnitude and orientation into a Gabor-like pattern. The inverse mapping recovers an
724 estimate by spectral peak detection with centroid refinement, followed by a consistency
725 step that combines extracted magnitude and angle to reconstruct the most likely log
726 value.